

Semianalytic Reconstruction of the Mean-Field Gain Model

and the Spectral Fluctuation Theory for ^{85}Rb Double- Λ Four-Wave Mixing

Separation of semiclassical gain, quantum fluctuations, and external detection loss

GABES model-audit report

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Scope and status. This document separates a semiclassical mean-field gain model, a quantum-fluctuation propagation model, and an external detection model. The mean-field chain and the quantum fluctuation *structure* are derived in closed form and verified numerically. A separate slow reference now solves the self-consistent pump-only state and infinitesimal trace-zero Nambu response for the standard minus Raman branch. It is not substituted into the production finite-seed gain path. A production frequency-dependent drift $M_q(\Omega_{\text{SA}})$ coupled to a microscopic atomic diffusion matrix $D(\Omega_{\text{SA}})$ is still not implemented, so no physically validated squeezing spectrum is claimed here. Any squeezing number obtained from the mean-field gains G_p and G_c alone is a diagnostic, not a prediction: the associated two-mode map fails the canonical commutator test ($\mathcal{G}_{\text{flux}} = G_p - (\omega_p/\omega_c)G_c \neq 1$) in the dissipative cell. Gains here follow the model's two-gain convention, in which G_p and G_c are computed independently against the same probe-seed input; the photon-flux gap, not the raw power-gain difference, is the canonical commutator diagnostic. The present claim gate is therefore: algebraic identities may pass, quantitative apparatus-level gain is UNSUPPORTED, and physical squeezing is UNAVAILABLE until the missing provenance, covariance, and held-out validation gates are closed.

Abstract

The report reconstructs the warm-vapor ^{85}Rb D1 double- Λ four-wave-mixing calculation in three explicitly separated layers: (i) a semiclassical atomic-response and mean-field gain model, (ii) a quantum-fluctuation propagation model with Langevin reservoirs, and (iii) an external detection model. For the standard minus Raman branch, a separate executable reference solves the pump-only state as a finite-dimensional Liouvillian null-space problem and resolves weak fluctuations at an independent analysis (sideband) frequency $\Omega_{\text{SA}}/2\pi = 0\text{--}4\text{ MHz}$. The production gain path remains the scan-converged finite-seed Floquet solve. The non-collinear two-photon Doppler shift is derived and evaluated in a separate pump-only reference, but is not yet included in either finite-seed production branch. The physical susceptibility is dimensionless while the reduced response has units of time; the Voigt prefactor, conjugate energy relation, and all 2π conventions are fixed explicitly.

Two analytic layers carry the frequency dependence. First, when the finite Liouvillian is diagonalizable, the weak-field resolvent has an *exact* simple-pole representation: the atomic response is a sum of complex Lorentzian terms whose isolated-pole centers are the *negative* imaginary parts and whose half-widths are the negative real parts of the Liouvillian eigenvalues. A defective Liouvillian instead requires a Jordan or contour representation and generally produces higher-order poles. Complete positivity places an exact GKSL generator in the closed stable half plane while still allowing peripheral modes. The numerical spectral check is therefore an assembly invariant, not an independent validation of model quality. Second, the distributed-noise covariance integral is solved as a differential Lyapunov equation, whose solution is an entire function of the Lyapunov

superoperator. Thus its finite- L coordinate-free evaluation has no algebraic singularity at an exceptional point, although the mode coalescence and parameter sensitivity of an exceptional point can still be physical. Resolving the solution in the eigenbasis exhibits the same trace-discriminant parametrization as the classical transfer matrix, while a block-exponential evaluation avoids the ill-conditioned eigenbasis entirely. Both results are verified against direct numerics at the 10^{-12} level or better in their stated fixtures.

Applying the pole representation to the documented operating point locates the slowest non-vanishing atomic pole at 0.520 MHz half-width at zero velocity, about 5.1 times the assumed ground-coherence rate, with per-class half-widths of 0.22–1.49 MHz across the velocity distribution. The non-collinear Doppler scale is instead a Gaussian rms width of about 1.38 MHz; it cannot be added in quadrature to those Lorentzian half-widths. A quantitative comparison with the measured 3.5 MHz squeezing bandwidth still awaits $S_-(\Omega_{SA})$. A separate slow pump-only reference now performs the two-dimensional Maxwell average: at 121 °C and 0.32° it gives 1.380174 MHz analytic rms and 1.380163 MHz by quadrature, while the finite-seed production scan remains one-dimensional.

A Floquet audit finds that $N_F = 1$ is not converged at the archived model point: going to $N_F = 2$ changes G_p by 16.8%, changes the archived conjugate transfer coefficient by 13.0%, and reverses the sign of the corresponding historical coefficient gap; $N_F = 2$ and $N_F = 3$ agree. The production core and compiled susceptibility kernel now implement the same finite-order recurrence for arbitrary $N_F \geq 1$. Reported seeded scans default to $N_F = 3$ and are accepted only after a full-scan comparison with $N_F = 2$ of the complex response and transfer coefficients, gains, wrapped phases, and the probe-gain optimum. The independent pump reference is gauge-equivalent to the minus-branch pump-only Floquet state to 8.37×10^{-15} . Across a declared 4×5 analysis-frequency/detuning scan on the full one-dimensional Doppler grid, its maximum normalized response residual is 7.32×10^{-17} . The finite-seed response approaches this reference monotonically as the seed is reduced, but the fitted last-three-point Rabi-amplitude slope is 1.749 and is not reported as an exact quadratic law. Its opt-in (v_z, v_x) wrapper keeps the laboratory beat and analysis frequency fixed, shifts only the atomic detunings, and passes the declared grid and cutoff convergence gates. Positivity of the truncated state is shown *not* to detect this error, so Floquet convergence testing cannot be replaced by a physicality check. The literature operating point $\Delta/2\pi \simeq +0.9$ GHz, $\delta/2\pi \simeq -8$ MHz, and $T = 121$ °C is kept distinct from the archived reduced-model point $(\Delta/2\pi, \delta/2\pi, T) = (-1.50$ GHz, -280 MHz, 110 °C). Agreement with the production code is classified only as an algebraic and numerical consistency check within shared assumptions.

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Part I

Definitions and conventions

1 Atomic references and frequency definitions

The four hyperfine states are

$$g_1 = 5S_{1/2}(F = 2), \quad g_2 = 5S_{1/2}(F = 3), \quad e_2 = 5P_{1/2}(F' = 2), \quad e_3 = 5P_{1/2}(F' = 3).$$

The optical reference is

$$\omega_{13} \equiv (E_{e_3} - E_{g_1})/\hbar, \quad \omega_{\text{HF}} \equiv (E_{g_2} - E_{g_1})/\hbar.$$

Every detuning and linewidth in an equation is an *angular* frequency. Numerical values are always displayed after division by 2π :

$$\frac{\Delta}{2\pi} = -1.50 \text{ GHz}, \quad \frac{\delta}{2\pi} = -280 \text{ MHz}, \quad \frac{\Gamma}{2\pi} = 5.746 \text{ MHz}.$$

The pump and probe-seed conventions are

$$\Delta = \omega_{\text{pump}} - \omega_{13}, \tag{1}$$

$$\omega_p = \omega_{\text{pump}} - \omega_{\text{HF}} + \delta, \tag{2}$$

$$\Omega_{\text{beat}} = \omega_{\text{pump}} - \omega_p = \omega_{\text{HF}} - \delta, \tag{3}$$

$$\omega_c = 2\omega_{\text{pump}} - \omega_p. \tag{4}$$

Thus Ω_{beat} is the optical pump–probe-seed beat that organizes the Floquet harmonics. It is not the analysis frequency.

The independent radio-frequency analysis variable is

$$\frac{\Omega_{\text{SA}}}{2\pi} \simeq 0.1\text{--}4 \text{ MHz}.$$

It labels quantum sidebands at $\omega_p \pm \Omega_{\text{SA}}$ and $\omega_c \mp \Omega_{\text{SA}}$. With the Fourier convention used here,

$$\delta \hat{a}_j(z, t) = \int_{-\infty}^{\infty} \frac{d\Omega_{\text{SA}}}{2\pi} \delta \hat{a}_j(z, \Omega_{\text{SA}}) e^{-i\Omega_{\text{SA}} t}. \tag{5}$$

Table 1: Frequency definitions used throughout the report.

Symbol	Definition	Reported as	Role
ω_{13}	$(E_{e_3} - E_{g_1})/\hbar$	$\omega_{13}/2\pi$	atomic optical reference
Δ	$\omega_{\text{pump}} - \omega_{13}$	$\Delta/2\pi$	pump one-photon detuning
δ	$\omega_p - \omega_{\text{pump}} + \omega_{\text{HF}}$	$\delta/2\pi$	Raman detuning
Ω_{beat}	$\omega_{\text{pump}} - \omega_p$	$\Omega_{\text{beat}}/2\pi$	optical pump–probe-seed beat
Ω_{SA}	sideband offset from an optical carrier frequency	$\Omega_{\text{SA}}/2\pi$	RF analysis (sideband)
ω_p, ω_c	probe and conjugate carrier angular frequencies	$\omega_{p,c}/2\pi$	optical carriers with $\omega_c = 2\omega_{\text{pump}} - \omega_p$
$-\text{Im } \lambda_m$	isolated-pole center in the convention of Eq. (5)	$-\text{Im } \lambda_m/2\pi$	compared with $\Omega_{\text{SA}}/2\pi$, never with $\Omega_{\text{beat}}/2\pi$
Γ	population-decay angular rate	$\Gamma/2\pi$	natural linewidth

Mode labels. The three optical fields are the *pump*, the *probe* (Stokes), and the *conjugate* (anti-Stokes); the weak field injected on the probe leg is the *probe-seed*, and probe and conjugate together are the *twin beams* [1]. Subscripts follow that naming: *p* denotes the probe and *c* the conjugate, while pump quantities carry the explicit subscript "pump" so that no symbol is overloaded. Powers are P_{pump} , P_0 for the injected probe-seed, and P_p , P_c for the two output beams. Macroscopic polarizations are written $\mathcal{P}_j$ to keep them distinct from powers, and the quadratures of mode *j* are (X_j, Y_j) .

2 Field, beam, and susceptibility normalization

Canonical traveling-wave amplitudes are photon-flux normalized:

$$[\delta\hat{a}_j(z, t), \delta\hat{a}_k^\dagger(z, t')] = \delta_{jk}\delta(t - t'), \quad P_j = \hbar\omega_j|\alpha_j|^2. \quad (6)$$

If the atomic code uses Rabi amplitudes $\Omega_j = \mathcal{G}_j\alpha_j$, its propagation matrix must be transformed before being interpreted quantum mechanically:

$$M_{\text{can}} = \mathcal{G}^{-1}M_\Omega\mathcal{G}, \quad \mathcal{G} = \text{diag}(\mathcal{G}_p, \mathcal{G}_c^*). \quad (7)$$

The near equality of the two optical wavelengths does not remove this normalization requirement. Every quantity below that is contracted with the canonical metric $J = \text{diag}(1, -1)$ — in particular the commutator-defect matrix of Sec. 16 — is basis dependent through Eq. (7) and must be recomputed if the collected mode areas differ.

The quoted beam sizes

$$w_{\text{pump}} = 530 \mu\text{m}, \quad w_p = 330 \mu\text{m}$$

are $1/e^2$ *intensity radii*, not diameters.

The reduced response is defined schematically by $\bar{\chi}_{jk} = \rho_{eg}^{(j \leftarrow k)}/\Omega_k$. Consequently

$$[\bar{\chi}_{jk}] = \text{s}, \quad \chi_{jk} = -\frac{2Nd^2}{\epsilon_0\hbar}\ell_{\text{norm}}\bar{\chi}_{jk}, \quad [\chi_{jk}] = 1. \quad (8)$$

The factor ℓ_{norm} is dimensionless. Its decomposition is audited in Appendix E. The structural factor is now $1/[2(2I+1)] = 1/12$: the trace-normalized ρ_{ss} supplies the ground-manifold population, while the inherited 0.74 remains an empirical residual rather than a first-principles quantity.

3 Operating-point classification

Table 2: Operating points are classified by provenance, not by numerical appearance in a scan.

Point and class	$\Delta/2\pi$	$\delta/2\pi$	T	Meaning
Archived reduced-model optimum	−1.50 GHz	−280 MHz	110 °C	Numerical optimum of the gain-only squeezing objective. It is not an experimental optimum and its $N_F = 1$ result is unconverged.
Sim et al. literature benchmark [1]	+0.9 GHz	−8 MHz	121 °C	Experimentally measured operating point: 600 mW pump, 8–10 μW probe-seed, 12.5 mm cell, $w_{\text{pump}} = 530 \mu\text{m}$, $w_p = 330 \mu\text{m}$, and $\theta = 0.32^\circ$.

The literature describes the gain as about 15; the cited ⁸⁵Rb squeezing trace reports raw powers $(P_0, P_p, P_c) = (8, 111, 109) \mu\text{W}$, which give the power ratios 13.875 and 13.625 used here. It reports maximum IDS of −7.8 dB, an approximate 3.5 MHz bandwidth, and a laser-source electronic-noise peak near 2 MHz. Those measurements are not properties of the archived $(\Delta/2\pi, \delta/2\pi) = (-1.50 \text{ GHz}, -280 \text{ MHz})$ model point.

Part II

Pump-only atomic steady state

4 Self-consistent finite-dimensional Liouvillian

For each velocity class and longitudinal position, the pump-only state is defined by

$$\mathcal{L}_{\text{pump}}(z, \mathbf{v})\rho_{\text{ss}}(z, \mathbf{v}) = 0, \quad \text{Tr } \rho_{\text{ss}}(z, \mathbf{v}) = 1. \quad (9)$$

For the standard minus Raman branch, the seeded frame has $\Omega_{\text{beat}} = \omega_{\text{HF}} - \delta$. The explicit transformation

$$U_{-}(t) = \exp(+i\Omega_{\text{beat}}t|g_2\rangle\langle g_2|) \quad (10)$$

is used with $\rho_{\text{pump}} = U_{-}^{\dagger}\rho_{\text{seed}}U_{-}$ and $H_{\text{pump}} = U_{-}^{\dagger}H_{\text{seed}}U_{-} - i\hbar U_{-}^{\dagger}\dot{U}_{-}$. It makes the second pump coupling static and changes the g_2 diagonal from δ to $\delta + \Omega_{\text{beat}} = \omega_{\text{HF}}$. The resulting pump Hamiltonian is

$$\frac{H_{\text{pump}}}{\hbar} = \omega_{\text{HF}}|g_2\rangle\langle g_2| - \sum_e \Delta_e(\mathbf{v})|e\rangle\langle e| + \frac{1}{2} \sum_{g,e} (\Omega_{ge}^{\text{pump}}|e\rangle\langle g| + \Omega_{ge}^{\text{pump}*}|g\rangle\langle e|), \quad (11)$$

where $\Delta_e(\mathbf{v}) = \Delta_e - \mathbf{k}_{\text{pump}} \cdot \mathbf{v}$. Spontaneous emission, collisional optical dephasing, and ground-state decoherence enter the same reference Liouvillian. The shared dissipator now includes the trace-preserving thermal reload channel $\gamma_t(\rho_{\text{th}} \text{Tr } \rho - \rho)$ with ground-manifold weights 5/12 and 7/12; the density-dependent Rb–Rb term remains a separate pure-dephasing channel. Vectorizing the 4×4 density matrix gives a 16×16 null-space problem; one row is replaced by the trace constraint. Solving that system makes populations and coherences feed back on one another self-consistently.

The diagonal phase in Eq. (10) leaves the declared projector dephasing and thermal reload invariant; phases acquired by individual collapse operators cancel in their Lindblad dissipators. The executable gauge map reconstructs the seeded-frame pump state with g_2 -row coherences in $n = -1$, g_2 -column coherences in $n = +1$, all remaining entries in $n = 0$, and all $|n| \geq 2$ harmonics exactly zero. The reconstructed state agrees with the direct $N_F = 3$ pump-only Floquet solution to 8.37×10^{-15} . This certification currently applies only to the standard minus Raman branch. The inherited plus-branch finite-seed frame fails this gauge-parity test and the pump-only reference therefore returns **unsupported** rather than mapping it silently.

Equations (9)–(11) are now an executable *semianalytic pump-only steady-state reference*: the Liouvillian matrix elements are analytic functions of the parameters and the null space is solved numerically. It remains separate from the current gain tables. The production path instead solves a trace-one, finite-seed density matrix at $N_F \geq 3$ and applies an adjacent-order full-scan convergence gate. The generalized audit separately repeats the same finite-seed finite-Floquet construction for $N_F = 1, 2, 3$. The older rate/Sylvester carrier in `analysis/squeezing/analytic_reconstruction/theory_final.py` is a third, historical leading-order approximation; it is not the source of the corrected Option-A tables. None of these distinctions changes the fact that any solve is exact only within its four-level, Markov, rotating-wave, field, velocity, and Floquet assumptions. In particular, the reference is undepleted, uses the one-dimensional longitudinal Doppler grid by default, and does not recompute the pump state segment by segment. Section 8 adds a separate two-dimensional wrapper around this reference; it does not promote that wrapper into the finite-seed production path.

The finite-seed path takes the probe-seed power already present at the cell as an input. It does not model the apparatus EOM V_{π} , insertion loss, RF-to- sideband conversion, or sideband impurity. Those quantities belong to the lab-to-cell input map and technical-noise budget, not to the pump-nullspace equation.

5 Validity of the historical rate/Sylvester factorization

An alternative construction solves saturation rate equations for populations and then solves a Sylvester equation for coherences. Because the resulting coherences do not feed back into the populations, that construction is not an exact strong-pump steady state. It may be used only under all of the following assumptions:

- optical coherences can be adiabatically eliminated;
- excited-state interference and coherent population trapping are small;
- pump-induced level dressing is weak compared with relevant separations;
- population redistribution is slow compared with optical dephasing; and
- fitted rates represent the omitted coherent dynamics.

At the archived point the quoted pump Rabi rate is comparable to or larger than the excited-state hyperfine splitting ($\Omega_{\text{pump}}/2\pi = 0.708$ GHz versus $\nu_{\text{HF}}^{(e)} = 361.58$ MHz), so these assumptions are not automatically satisfied. All population and coherence numbers obtained this way are therefore classified as a controlled approximation only if a separate error study is supplied.

6 Phenomenological rates and truncations

The four-level pump model uses a Zeeman-averaged hyperfine basis. The ground-coherence floor, density-dependent collisional term, and optical self-broadening are phenomenological inputs unless independently measured. At the literature point the engine evaluates

$$\frac{\gamma_{gg}}{2\pi} = 102.882 \text{ kHz}, \quad \frac{\gamma_{\text{opt}}}{2\pi} = 0.7416 \text{ MHz}, \quad \frac{\Gamma}{2\pi} = 5.7460 \text{ MHz}, \quad (12)$$

for $N = 2.1496 \times 10^{19} \text{ m}^{-3}$ at $T = 121^\circ\text{C}$, where the effective ground-coherence decay is the sum of the 100 kHz trace-preserving transit reset and a 2.882 kHz Rb–Rb pure-dephasing term, and $\gamma_{\text{opt}} = \frac{1}{2}\beta N$ is the added collisional optical half-width. Thus the transit contribution replenishes ground populations rather than merely damping the Raman coherence, while preserving trace exactly. Their provenance and sensitivity are listed in Appendix D. No calculation involving these quantities is labeled exact.

Part III

Weak-field linear response

7 Frequency-dependent response matrix

Let $\mathcal{V}_k(\Omega_{\text{SA}})$ be the perturbation generated by the weak probe or conjugate sideband, and let $\omega_k(\delta, \Omega_{\text{SA}})$ be that field's angular frequency in the static pump frame. With Eqs. (5) and $\dot{\rho} = \mathcal{L}_{\text{pump}}\rho + \mathcal{V}\rho_{\text{ss}}$,

$$\delta\rho_k(\Omega_{\text{SA}}) = -(\mathcal{L}_{\text{pump}} + i\omega_k(\delta, \Omega_{\text{SA}}))_{\text{Tr}=0}^{-1} \mathcal{V}_k(\Omega_{\text{SA}})\rho_{\text{ss}}. \quad (13)$$

For the implemented minus-branch Nambu sector,

$$\omega_k^{(-)}(\delta, \Omega_{\text{SA}}) = -\omega_{\text{HF}} + \delta + \Omega_{\text{SA}}. \quad (14)$$

Thus the optical two-photon detuning and RF analysis frequency are independent inputs; changing one never silently changes the other. The inverse acts on the trace-zero subspace. At $\omega_k = 0$ the implementation projects the full stationary subspace instead of applying an ordinary singular inverse. This zero relative resolvent frequency is not generally the same as setting the lab analysis frequency

Ω_{SA} to zero. Using the opposite Fourier sign would reverse the sign of the imaginary term; mixing those conventions is not allowed.

Polarization readout gives the full response matrix

$$\begin{pmatrix} \mathcal{P}_p \\ \mathcal{P}_c^* \end{pmatrix} = \epsilon_0 \begin{pmatrix} \chi_{pp}(\Omega_{\text{SA}}) & \chi_{pc}(\Omega_{\text{SA}}) \\ \chi_{cp}^*(\Omega_{\text{SA}}) & \chi_{cc}^*(\Omega_{\text{SA}}) \end{pmatrix} \begin{pmatrix} E_p \\ E_c^* \end{pmatrix}. \quad (15)$$

Thus M_q , the reservoir coupling B , and the diffusion matrix D may all depend on Ω_{SA} . The executable reference returns the complete complex 2×2 derivative with Nambu input/output order (p, c^*) and units of seconds per unit angular Rabi frequency. It supplies the atomic-response ingredient needed for a future $M_q(\Omega_{\text{SA}})$ construction, but it does not construct B or D and is not the production gain solver.

8 Non-collinear Doppler geometry

Choose the pump along z and the probe-seed in the x - z plane:

$$\mathbf{k}_{\text{pump}} = k_{\text{pump}} \hat{\mathbf{z}}, \quad \mathbf{k}_p = k_p (\sin \theta \hat{\mathbf{x}} + \cos \theta \hat{\mathbf{z}}).$$

The laboratory detunings seen by an atom with velocity $\mathbf{v} = (v_x, v_y, v_z)$ are therefore

$$\Delta_{\text{eff}} = \Delta - k_{\text{pump}} v_z, \quad (16)$$

$$\delta_{\text{eff}} = \delta + (k_{\text{pump}} - k_p \cos \theta) v_z - k_p \sin \theta v_x. \quad (17)$$

The sign of the transverse term follows directly from $\delta_{\text{eff}} = \delta - (\mathbf{k}_p - \mathbf{k}_{\text{pump}}) \cdot \mathbf{v}$. The quantitative average is at least two-dimensional:

$$\langle X \rangle_v = \iint \frac{dv_z dv_x}{2\pi\sigma_v^2} \exp\left[-\frac{v_z^2 + v_x^2}{2\sigma_v^2}\right] X(\Delta_{\text{eff}}, \delta_{\text{eff}}). \quad (18)$$

The unused v_y projection integrates to one. For $k_{\text{pump}} \simeq k_p = k$,

$$|\mathbf{k}_{\text{pump}} - \mathbf{k}_p| = 2k \sin(\theta/2) \simeq k\theta, \quad \frac{\sigma_{\delta,\theta}}{2\pi} \simeq \frac{\theta\sigma_v}{\lambda} = 1.361 \text{ MHz} \quad (19)$$

for $\theta = 0.32^\circ$, $\sigma_v = 193.7 \text{ m/s}$, and $\lambda = 795 \text{ nm}$. These values correspond to the 110°C estimate. At the 121°C literature point, $\sigma_v \simeq 196.5 \text{ m/s}$ and the width is about 1.38 MHz . Both are much larger than $\gamma_{gg}/2\pi = 0.103 \text{ MHz}$. The angular term vanishes as $\theta \rightarrow 0$; a roughly 2 kHz collinear residual remains because $\omega_{\text{pump}} \neq \omega_p$.

The minus-branch pump-only reference now implements Eq. (18) as a tensor Gauss–Legendre quadrature of the truncated Maxwell distribution. Its interface keeps $\Omega_{\text{beat,lab}}$ and Ω_{SA} independent of velocity and applies Eq. (17) only to the atomic detunings. This separation is essential: passing δ_{eff} to the production Floquet table would also move the laboratory beat frequency and would describe a different problem.

At the literature point the analytic rms is 1.380173564 MHz , of which 1.380161171 MHz is transverse and 5.848714 kHz is axial at angle. The 40×40 , $5\sigma_v$ quadrature gives 1.380163304 MHz , a $-7.43 \times 10^{-4}\%$ difference. Table 3 records the feature-level convergence. These large gains are mean-field diagnostics under the same uncalibrated coupling assumptions as the other gain tables, not experimental predictions.

The accepted successive-grid comparison is $32 \rightarrow 40$: the gain changes by 0.00711% and the sampled feature position by 0.05 MHz . Extending the cutoff from $4.5\sigma_v$ to $5\sigma_v$ changes the gain by 0.05215% and does not move the sampled feature. At the declared -8 MHz literature point, the one-dimensional and two-dimensional diagnostics are respectively $G_p = 403.825150$ and 404.304420 , a $+0.118683\%$ change. The response residual is 2.18×10^{-16} and the trace error is 5.83×10^{-26} .

This closes the slow-reference quadrature task, not the production geometry: the finite-seed scan remains one-dimensional, the crossing angle is a single value rather than a beam angular distribution, and the pump state is not recomputed segment by segment. No microscopic noise is added.

Table 3: Two-dimensional non-collinear reference convergence. The refined scan spacing is 0.05 MHz; the first three rows use a $5\sigma_v$ cutoff.

Grid or cutoff	velocity pairs	feature $\delta/2\pi$	G_p	change
24×24	576	36.15 MHz	4517.4025	+0.01495% vs 40
32×32	1024	36.00 MHz	4517.0481	+0.00711% vs 40
40×40	1600	35.95 MHz	4516.7271	reference
$4.5\sigma_v$ at 40	1600	35.95 MHz	4519.0828	+0.05215%
$5.0\sigma_v$ at 40	1600	35.95 MHz	4516.7271	reference

9 Floquet harmonics and convergence requirement

If the chosen rotating frame leaves a periodic pump term at Ω_{beat} , expand

$$\rho(t) = \sum_{n=-N_F}^{N_F} \rho_n e^{-in\Omega_{\text{beat}}t}.$$

The finite block equations are

$$(\mathcal{L}_0 + in\Omega_{\text{beat}}) \rho_n + \mathcal{C}_p \rho_{n-1} + \mathcal{C}_m \rho_{n+1} = 0. \quad (20)$$

Trace one is imposed in the $n = 0$ sector and trace zero in every other sector. Before truncation, the same blocks define an infinite-dimensional Floquet–Liouville operator $\mathbb{L}$; its eigenvalues are Floquet exponents, defined modulo $i\Omega_{\text{beat}}$. When the relevant spectral decomposition exists, the resolvent identities of Sec. 10 then apply on that extended space, with every harmonic sector contributing a frequency replica; a simple-pole sum additionally requires diagonalizability.

For the exact periodic GKSL problem, the one-period propagator is CPTP, so each Floquet multiplier $r_m = \exp(\lambda_m T)$ obeys $|r_m| \leq 1$. A finite- N_F projected block matrix is not generally the generator or logarithm of a CPTP map, however; positive real parts may be truncation artifacts, and passing a half-plane or state-positivity test does not establish convergence. The truncation is accepted only after convergence with respect to N_F has been established for the transfer coefficients, their wrapped phases, and the specified optimum δ^* . The production gate uses the dimensionless mixed tolerance $|\Delta T| \leq 10^{-10} + 0.01 \max(|T_N|, |T_{N-1}|)$ for every transfer coefficient. Reduced susceptibilities have units of time, so their absolute floor is instead 10^{-8} times that component’s full-scan maximum; this is added to the same one-percent local relative term. Derived gains must change by less than one percent, retained wrapped phases by less than 0.5° , and the probe-gain optimum by at most one final-grid interval. A failed condition returns **UNCONVERGED**. The production core and compiled kernel now use generalized block continued fractions and are pinned for $N_F = 1, 2, 3$ against a separately assembled dense finite block, including trace and normalized-equation residual tests. The archived part of the audit in Sec. 25 retains its own assembly, reproduces `ref_solver.py`’s historical $N_F = 1$ finite-seed solve, and agrees with direct finite-block solves to 5.4×10^{-13} . Those numbers isolate truncation under the historical dressed- k convention; they are not production Option-A physical predictions. The separate literature-point section uses the current density-matrix assembly and checks only its Maxwell/ Q propagation against a literal independent SI reconstruction. The separate pump-only reference implements Eq. (9) for the minus branch and is checked against both its gauge-mapped finite-Floquet state and an independent dense/direct response construction; it is not substituted into either of those finite-seed paths.

10 Exact spectral representation of the resolvent

Equation (13) is usually evaluated by a numerical matrix inversion at each Ω_{SA} , which hides the analytic structure of the response. The representation below exposes it without approximation.

10.1 Eigenbasis of the Liouvillian

Let $\mathcal{L} \equiv \mathcal{L}_{\text{pump}}$ act on the n^2 -dimensional vectorized space and let its right and left eigenvectors be

$$\mathcal{L}|r_m\rangle = \lambda_m|r_m\rangle, \quad \langle l_m|\mathcal{L} = \lambda_m\langle l_m|, \quad \langle l_m|r_{m'}\rangle = \delta_{mm'}, \quad (21)$$

where $\langle \cdot | \cdot \rangle$ is the Hilbert–Schmidt pairing. $\mathcal{L}$ is not Hermitian, so left and right eigenvectors differ and the eigenvalues are complex. Whenever $\mathcal{L}$ is diagonalizable, the resolvent has the exact partial-fraction form

$$(\mathcal{L} + i\Omega)^{-1} = \sum_m \frac{|r_m\rangle\langle l_m|}{\lambda_m + i\Omega}. \quad (22)$$

Substituting into Eq. (13) and contracting with the Clebsch–Gordan-weighted polarization readout $\langle w_j|$ gives the reduced response as a pole sum,

$$\bar{\chi}_{jk}(\Omega) = \sum_m \frac{A_m^{(jk)}}{\lambda_m + i\Omega}, \quad A_m^{(jk)} = -\langle w_j|r_m\rangle\langle l_m|\mathcal{V}_k|\rho_{\text{ss}}\rangle. \quad (23)$$

Equation (23) is an identity, not a truncation. It is a similarity transform of the same linear system, so it carries no approximation beyond diagonalizability of $\mathcal{L}$.

For a defective eigenvalue, degeneracy alone is not the issue: a Jordan block $J_\lambda = \lambda I + N$ of size p , with $N^p = 0$, contributes

$$(J_\lambda + i\Omega I)^{-1} = \sum_{r=0}^{p-1} \frac{(-1)^r N^r}{(\lambda + i\Omega)^{r+1}}. \quad (24)$$

Thus the resolvent identity survives, but the simple Lorentzian pole sum does not: defective blocks generate higher-order poles and should be evaluated by a Jordan-free linear solve or contour/Riesz-projector construction in numerical work.

10.2 The trace-mode residue and the zero-frequency inverse

$\mathcal{L}$ always has the eigenvalue $\lambda_0 = 0$, whose right eigenvector is the steady state ρ_{ss} . Its term in Eq. (22) diverges as $\Omega \rightarrow 0$. For a unique stationary state and the traceless commutator drive used here, its trace-mode residue vanishes; the zero-frequency inverse still requires an explicit trace-zero or Drazin construction.

Trace preservation means $\text{Tr}[\mathcal{L}[X]] = 0$ for every X , which says precisely that the trace functional is the *left* eigenvector belonging to λ_0 :

$$\langle l_0| = \text{Tr}(\cdot). \quad (25)$$

Assume first that the pump Liouvillian has a unique stationary state. The drive is a commutator, $\mathcal{V}_k\rho_{\text{ss}} = -i[V_k, \rho_{\text{ss}}]$, and is therefore traceless. Hence

$$A_0^{(jk)} \propto \langle l_0|\mathcal{V}_k|\rho_{\text{ss}}\rangle = \text{Tr}(-i[V_k, \rho_{\text{ss}}]) = 0 \quad (26)$$

identically, for any V_k and any ρ_{ss} . For $\Omega \neq 0$ the stationary-mode contribution to the contracted response therefore vanishes. At $\Omega = 0$, however, the full inverse does not exist and the formal stationary term is $0/0$; one must retain the trace-zero solve of Eq. (13), omit the stationary subspace explicitly, or use the Drazin inverse. If the Liouvillian has more than one stationary mode, tracelessness alone does not eliminate the residues of every zero mode, and the whole stationary subspace must be projected. Numerically, the trace-mode residue vanishes exactly for the engine’s Liouvillian and drives.

10.3 Lorentzian reading

Write the eigenvalues as

$$\lambda_m = -\Gamma_m + i\omega_m, \quad \Gamma_m \geq 0, \quad (27)$$

and absorb the sign into the residue, $B_m \equiv -A_m$. In the diagonalizable simple-pole case, each term of Eq. (23) is then a complex Lorentzian in standard form,

$$\bar{\chi}_{jk}(\Omega) = \sum_m \frac{B_m^{(jk)}}{\Gamma_m - i(\Omega + \omega_m)}, \quad (28)$$

For an isolated pole with a slowly varying residue, the amplitude-squared profile has HWHM Γ_m and center $\Omega = -\omega_m = -\text{Im } \lambda_m$. A complex residue mixes absorptive and dispersive components, and overlapping poles need not display a peak exactly at that center. Subject to those qualifications, the parametrization assigns a direct physical reading to the numerically computed spectrum of $\mathcal{L}$:

- $\text{Im } \lambda_m$ is a *dressed oscillation frequency*: a generalized Rabi frequency, an Autler–Townes or hyperfine splitting, or a Raman detuning, depending on which coherence dominates $|r_m\rangle$.
- $-\text{Re } \lambda_m$ is the corresponding *power-broadened relaxation rate*. It is not the bare Γ , γ_{gg} , or γ_{opt} ; strong driving mixes the channels and shifts the widths.
- The residues A_m decide which poles are actually visible in a given response element. A pole with small Γ_m contributes a narrow feature only if its residue is not also small.

10.4 Positivity of the pole half-widths, and what it does not prove

Because $\mathcal{L}$ is a Lindblad generator [8], the semigroup $e^{\mathcal{L}t}$ is completely positive and trace preserving for all $t \geq 0$. Its spectrum therefore satisfies

$$\text{Re } \lambda_m \leq 0 \quad \text{for every } m, \quad (29)$$

with at least one eigenvalue exactly zero, whose right eigenvector is the steady state. Two consequences follow.

First, Eq. (29) places the poles in the closed stable half plane. Finiteness on every real Ω additionally requires a unique relaxing steady state, or an explicit check that every nonstationary mode has $\text{Re } \lambda_m < 0$; complete positivity alone permits peripheral imaginary modes and decoherence-free sectors.

Second, Eq. (29) is guaranteed for an exactly assembled Lindblad generator. A numerical spectral check is therefore an implementation invariant rather than an independent validation of model quality: it can detect assembly or truncation mistakes, but passing it does not validate omitted physics. Direct evaluation confirms the invariant: at both operating points of Table 2 the largest real part found over the velocity grid is at most $+1.7 \times 10^{-6} \text{ s}^{-1}$, which is the null eigenvalue resolved to better than 5×10^{-14} relative to the scale $\Gamma = 3.6 \times 10^7 \text{ s}^{-1}$. Model errors — including the Floquet truncation error of Sec. 25 — must be caught by convergence and comparison tests instead (Sec. 26).

10.5 Effect of the velocity average

Equation (23) holds *per velocity class*. The velocity average of Sec. 8 replaces the discrete pole set by a distribution: each $\lambda_m(\mathbf{v})$ traces a curve as $\mathbf{v}$ varies, and the averaged response is an integral over that curve rather than a finite sum of Lorentzians. This is the microscopic origin of inhomogeneous broadening; only the simple-pole limit below reduces to a Voigt profile.

The two pole families behave very differently, because the Doppler shift enters the excited-state diagonal as $\Delta_{\text{eff}} = \Delta - k_{\text{pump}} v_z$:

- **Optical poles** shift by the full one-photon rms scale $k\sigma_v/2\pi = \sigma_v/\lambda$, about 244 MHz at 110 °C and 247 MHz at 121 °C, and are inhomogeneously smeared out.

- **Ground-Raman poles** cancel that common axial shift in the collinear limit. On the one-dimensional axial grid used below, their center therefore stays near zero, while their *width* varies through velocity-dependent optical pumping; light shifts can produce residual center shifts. The separate full non-collinear reference additionally distributes their centers through the transverse term in Eq. (17); it does not produce the full optical smear.

Section 24 quantifies both statements at the literature operating point. The residual non-collinear distribution $\sigma_{\delta,\theta}$ of Eq. (19) convolves with the velocity-dependent Raman response. The separate reference performs this average, but a measured squeezing bandwidth additionally requires microscopic noise propagation and detector transfer.

11 Faddeeva base integral and Voigt identity

For a single velocity projection,

$$\left\langle \frac{1}{x_0 - kv + i\Gamma_o} \right\rangle_v = -\frac{i}{k\sigma_v} \sqrt{\frac{\pi}{2}} w\left(\frac{x_0 + i\Gamma_o}{\sqrt{2}k\sigma_v}\right), \quad w(z) = e^{-z^2} \operatorname{erfc}(-iz). \quad (30)$$

With

$$V(x; \sigma, \gamma) = \frac{\operatorname{Re} w[(x + i\gamma)/(\sqrt{2}\sigma)]}{\sqrt{2\pi}\sigma},$$

the dimensionally correct identity is

$$\boxed{\operatorname{Im} \left\langle \frac{1}{x_0 - kv + i\Gamma_o} \right\rangle_v = -\pi V(x_0; k\sigma_v, \Gamma_o).} \quad (31)$$

There is no additional factor $1/k$. Equation (30) is an exact base simple-pole integral, and it is precisely the velocity average of a *single* term of the pole sum (23) under the assumption that the pole position is linear in v and its residue is constant. Velocity-dependent populations, Raman self-energies, and the two-detuning geometry violate that assumption and generally require multidimensional quadrature or a justified partial-fraction reduction. A one-dimensional Faddeeva evaluation is not a complete treatment of the experimental geometry.

Part IV

Semiclassical field propagation and gain

12 Classical coupled-mode equation

Use the GABES positive-frequency convention $E_j^{(+)} \propto \alpha_j(z) e^{ik_{j,z}z - i\omega_j t}$ and define the physical susceptibility so that $\operatorname{Im} \chi_{jj} > 0$ is passive absorption. For $\Delta k_{\text{geom}} = 2k_{\text{pump}} - k_p - k_c$, the symmetric mismatch gauge is $\tilde{\alpha}_p = \alpha_p \exp(-i\Delta k_{\text{geom}}z/2)$ and $\tilde{\alpha}_c^* = \alpha_c^* \exp(+i\Delta k_{\text{geom}}z/2)$. It gives diagonal phases $(-i\Delta k_{\text{geom}}/2, +i\Delta k_{\text{geom}}/2)$ in the two-component basis $(\tilde{\alpha}_p, \tilde{\alpha}_c^*)^T$. This fixes the code-level Maxwell convention directly; in particular a passive uncoupled probe obeys $|\alpha_p(L)|^2 = \exp[-k_p \operatorname{Im} \chi_{pp} L] |\alpha_p(0)|^2$. Direct differentiation then gives

$$\frac{d}{dz} \begin{pmatrix} \tilde{\alpha}_p \\ \tilde{\alpha}_c^* \end{pmatrix} = M_{\text{cl}} \begin{pmatrix} \tilde{\alpha}_p \\ \tilde{\alpha}_c^* \end{pmatrix}, \quad M_{\text{cl}} = Q^{-1} M_E Q, \quad (32)$$

where the Maxwell electric-field matrix is

$$M_E = \begin{pmatrix} +\frac{ik_p}{2} \chi_{pp} - \frac{i\Delta k_{\text{geom}}}{2} & +\frac{ik_p}{2} \chi_{pc} \\ -\frac{ik_c}{2} \chi_{cp}^* & -\frac{ik_c}{2} \chi_{cc}^* + \frac{i\Delta k_{\text{geom}}}{2} \end{pmatrix}, \quad Q = \operatorname{diag} \left(\sqrt{\frac{2\hbar\omega_p}{\epsilon_0 c A_p}}, \sqrt{\frac{2\hbar\omega_c}{\epsilon_0 c A_c}} \right). \quad (33)$$

Here

$$\Delta k_{\text{geom}} = 2k_{\text{pump},z} - k_{p,z} - k_{c,z} \quad (34)$$

uses vacuum carrier wave numbers and geometry only. The effective mode areas A_p, A_c must match the collected modes. Equation (32) is the canonical photon-flux rescaling of the electric-field equation; a Rabi-amplitude implementation must equivalently apply Eq. (7).

No double counting of dispersion

Equation (32) is Option A. The real parts of χ_{pp} and χ_{cc} generate dispersive phase exactly once through the diagonal propagation terms. Refractive indices $n_j = 1 + \text{Re } \chi_{jj}/2$ are *not* inserted again into Δk_{geom} . A dressed- k calculation that uses both mechanisms is retained only for implementation history and convergence diagnostics.

13 Constant-matrix exponential and mean-field gains

For an undepleted pump and z -independent coefficients,

$$\begin{aligned} T_{\text{cl}}(L) &= e^{M_{\text{cl}}L} \\ &= e^{sL} \left[\cosh(qL)I + \frac{\sinh(qL)}{q}(M_{\text{cl}} - sI) \right], \end{aligned} \quad (35)$$

$$s = \frac{a+d}{2}, \quad q = \sqrt{\frac{(a-d)^2}{4} + bc}, \quad (36)$$

where a, b, c, d are the entries of M_{cl} , so that M_{cl} has eigenvalues

$$\mu_{\pm} = s \pm q. \quad (37)$$

Equation (36) needs no eigenvectors. Split off the trace, $M_{\text{cl}} = sI + N$ with N traceless. For a traceless 2×2 matrix the Cayley–Hamilton theorem reads $N^2 = -\det(N)I = q^2I$, so the exponential series collapses into even and odd parts,

$$e^{NL} = \sum_{k \text{ even}} \frac{(qL)^k}{k!} I + \sum_{k \text{ odd}} \frac{(qL)^k}{k!} \frac{N}{q} = \cosh(qL)I + \frac{\sinh(qL)}{q}N, \quad (38)$$

and $e^{M_{\text{cl}}L} = e^{sL}e^{NL}$ because sI commutes with N . Since $\sinh(z)/z$ is entire, Eq. (36) is valid verbatim at $q = 0$, where it reads $e^{sL}(I + LN)$; no limit needs to be taken. The same 2×2 trace-discriminant parametrization reappears in the quantum-noise integral of Sec. 18. Its numerical pair generally differs because $M_q(\Omega)$ need not equal M_{cl} .

The formula is an algebraic identity for constant M_{cl} . Its physical assumptions are:

- undepleted pump;
- fixed density and temperature;
- no longitudinal pump attenuation;
- z -independent overlap and response coefficients; and
- a single collected transverse mode.

If those assumptions fail, use an ordered product or path-ordered exponential.

13.1 Gain convention

Both gains are referred to the incident probe-seed power P_0 and read from a single photon-flux transfer matrix evaluated with probe-seed-only input:

$$G_p \equiv \frac{P_p^{\text{out}}}{P_0} = |T_{\text{cl},11}|^2, \quad G_c \equiv \frac{P_c^{\text{out}}}{P_0} = \frac{\omega_c}{\omega_p} |T_{\text{cl},21}|^2. \quad (39)$$

This is a **two-gain convention**: G_p and G_c are computed independently. Two gaps must be distinguished:

$$\mathcal{G}_{\text{flux}} \equiv |T_{\text{cl},11}|^2 - |T_{\text{cl},21}|^2 = G_p - \frac{\omega_p}{\omega_c} G_c, \quad \mathcal{G}_{\text{power}} \equiv G_p - G_c. \quad (40)$$

Both are outputs rather than inputs. Canonical commutator preservation constrains $\mathcal{G}_{\text{flux}}$, which equals one in the closed lossless parametric limit of Eq. (43). The power gap is numerically close because $\omega_p \simeq \omega_c$, but it is not the canonical invariant.

The distinction matters because a near-degenerate single-gain power convention is also in use. The exact photon-flux relation is

$$P_p = G P_0, \quad P_c = \frac{\omega_c}{\omega_p} (G - 1) P_0. \quad (41)$$

The commonly quoted $P_c = (G - 1)P_0$ form additionally uses $\omega_c/\omega_p \simeq 1$. In either form the conjugate photon-flux gain has been imposed rather than computed, and $\mathcal{G}_{\text{flux}} = 1$ by construction. Equation (41) is the ideal-parametric special case of Eq. (39). The mapping between the two conventions is

$$G_p \longleftrightarrow G, \quad G_c \longleftrightarrow \frac{\omega_c}{\omega_p} (G - 1), \quad (42)$$

valid only where $\mathcal{G}_{\text{flux}} = 1$. Two consequences follow. A conjugate-to-probe-seed power ratio quoted under Eq. (41) is not an independently calibrated G_c , so it must not be compared with the computed G_c as though it were. The photon-flux gap also cannot serve as a validation target against a data set in which it was assumed.

The quantities in Eq. (39) are explicitly **semiclassical mean-field gains**. They do not by themselves determine a quantum covariance.

14 Classical photon-flux identity

For $u' = au + bv$ and $v' = cu + dv$,

$$\frac{d}{dz}(|u|^2 - |v|^2) = 2 \operatorname{Re}(a)|u|^2 - 2 \operatorname{Re}(d)|v|^2 + 2 \operatorname{Re}[(b - c^*)u^*v]. \quad (43)$$

In a closed lossless parametric model, $\operatorname{Re} a = \operatorname{Re} d = 0$ and $b = c^*$, so $\mathcal{G}_{\text{flux}} = 1$. Equation (43) is a property of the classical transfer matrix. If $\mathcal{G}_{\text{flux}} \neq 1$, it is evidence that a two-operator quantum map needs additional channels; it is not permission to use a noncanonical Bogoliubov map. Because the two-gain convention computes $\mathcal{G}_{\text{flux}}$ rather than imposing it, this test is available at every scan point.

Part V

Quantum fluctuation propagation

15 Quantum-Langevin propagation

Define the two-mode sideband vector in the Bogoliubov (phase-conjugate) pairing that the four-wave process couples,

$$\hat{\mathbf{A}}(z, \Omega) = \begin{pmatrix} \delta \hat{a}_p(z, \Omega) \\ \delta \hat{a}_c^\dagger(z, -\Omega) \end{pmatrix}, \quad [\hat{\mathbf{A}}(z, \Omega), \hat{\mathbf{A}}^\dagger(z, \Omega')] = 2\pi J \delta(\Omega - \Omega'), \quad J = \operatorname{diag}(1, -1). \quad (44)$$

The metric J is the one preserved by a two-mode Bogoliubov transformation; a lossless parametric amplifier acts on $\hat{\mathbf{A}}$ by an element of $SU(1, 1)$, and the requirement $TJT^\dagger = J$ used below is exactly

that condition. Equivalently, in the quadrature basis of Eq. (52) it is the symplectic condition. The distributed fluctuation equation is

$$\frac{\partial}{\partial z} \hat{\mathbf{A}}(z, \Omega_{\text{SA}}) = M_q(z, \Omega_{\text{SA}}) \hat{\mathbf{A}}(z, \Omega_{\text{SA}}) + B(z, \Omega_{\text{SA}}) \hat{\mathbf{F}}(z, \Omega_{\text{SA}}). \quad (45)$$

Its formal solution is

$$T(z_2, z_1; \Omega) = \mathcal{P} \exp \left[\int_{z_1}^{z_2} M_q(z, \Omega) dz \right], \quad (46)$$

$$\hat{\mathbf{A}}_{\text{out}}(\Omega) = T(L, 0; \Omega) \hat{\mathbf{A}}_{\text{in}}(\Omega) + \int_0^L T(L, z; \Omega) B(z, \Omega) \hat{\mathbf{F}}(z, \Omega) dz. \quad (47)$$

For local Markov reservoirs,

$$[\hat{\mathbf{F}}(z, \Omega), \hat{\mathbf{F}}^\dagger(z', \Omega')] = 2\pi J_F \delta(z - z') \delta(\Omega - \Omega').$$

16 Preservation of the commutation relations

The output operators must obey the same commutation relations as the input operators. With the reservoir contribution included this requires

$$\boxed{T J T^\dagger + \int_0^L T(L, z) B J_F B^\dagger T^\dagger(L, z) dz = J,} \quad (48)$$

which is the statement that gain and loss must each be accompanied by their own Langevin noise. A sufficient local condition is

$$\boxed{M_q J + J M_q^\dagger + B J_F B^\dagger = 0.} \quad (49)$$

Define the Hermitian commutator-defect matrix

$$K \equiv - \left(M_q J + J M_q^\dagger \right) = B J_F B^\dagger. \quad (50)$$

The signs of its eigenvalues classify the reservoir channels the drift requires: a positive eigenvalue of K admits a loss-type (vacuum-absorbing) channel with $J_F = +1$, while a negative eigenvalue requires a gain-type (inverted) channel with $J_F = -1$. Both are physical for a parametric amplifier; the sign pattern is a *channel classification*, not a validity test. Because K is defined through the canonical metric J , it is sensitive to the normalization of Eq. (7) and must be recomputed whenever the collected mode areas or Rabi-to-field factors change.

17 Covariance and atomic diffusion

Define reservoir symmetrized correlations by

$$\frac{1}{2} \left\langle \{ \hat{\mathbf{F}}(z, \Omega), \hat{\mathbf{F}}^\dagger(z', \Omega') \} \right\rangle = 2\pi N_{\text{res}}(z, \Omega) \delta(z - z') \delta(\Omega - \Omega'), \quad D = B N_{\text{res}} B^\dagger.$$

Then

$$\boxed{V_{\text{out}}(\Omega) = T(\Omega) V_{\text{in}}(\Omega) T^\dagger(\Omega) + W(L; \Omega), \quad W(L; \Omega) \equiv \int_0^L T(L, z; \Omega) D(z, \Omega) T^\dagger(L, z; \Omega) dz.} \quad (51)$$

Equation (51) is one 2×2 two-mode block. Before applying the detector model, solve its companion block and assemble

$$\hat{\Xi}(\Omega) = (\hat{a}_p(\Omega), \hat{a}_c(\Omega), \hat{a}_p^\dagger(-\Omega), \hat{a}_c^\dagger(-\Omega))^T$$

with spectral covariance $V_{\Xi}(\Omega)$. The quadrature-sideband covariance used below is

$$V_R(\Omega) = C V_{\Xi}(\Omega) C^{\dagger}, \quad C = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 & 0 \\ -i & 0 & i & 0 \\ 0 & 1 & 0 & 1 \\ 0 & -i & 0 & i \end{pmatrix}, \quad (52)$$

corresponding to $\hat{\mathbf{R}} = (X_p, Y_p, X_c, Y_c)^T$. This explicit assembly prevents the 2×2 two-mode covariance from being mixed dimensionally with the 4×4 detector matrices.

The commutator condition fixes antisymmetric reservoir correlations, not the symmetrized diffusion that determines squeezing. A physical $D(z, \Omega_{\text{SA}})$ must be derived from the pumped atomic Lindblad model, generalized Einstein relations, or an independently justified reservoir model. Independent velocity classes add as independent diffusion powers; one must not square a velocity-averaged noise amplitude.

18 Closed-form evaluation of the distributed-noise integral

The integral $W(L; \Omega)$ in Eq. (51) is the only part of the fluctuation chain that is not already an algebraic operation. For a z -independent drift it is available in closed form. Its 2×2 trace-discriminant structure parallels the mean-field transfer matrix, but it is formed from $M_q(\Omega)$ itself.

18.1 Differential Lyapunov form

For constant M_q the propagator depends only on the separation, $T(L, z) = e^{M_q(L-z)}$, so substituting $u = L - z$ gives

$$W(L) = \int_0^L e^{M_q u} D e^{M_q^{\dagger} u} du. \quad (53)$$

Introduce the *Lyapunov superoperator*, a linear map on matrices,

$$\mathcal{M}[X] \equiv M_q X + X M_q^{\dagger}, \quad \text{so that} \quad e^{u\mathcal{M}}[D] = e^{M_q u} D e^{M_q^{\dagger} u}. \quad (54)$$

The second identity holds because left and right multiplication commute. Equation (53) is then simply $\int_0^L e^{u\mathcal{M}}[D] du$, and differentiating with respect to the upper limit gives the differential Lyapunov equation

$$\boxed{\frac{dW}{dL} = \mathcal{M}[W] + D, \quad W(0) = 0.} \quad (55)$$

W is Hermitian for Hermitian D , because $\mathcal{M}$ preserves Hermiticity, and positive semidefinite for positive semidefinite D , because the integrand of Eq. (53) is a congruence of D . Both are useful numerical invariants: a computed W that violates either signals an implementation error rather than a physical effect.

18.2 Solution

Equation (55) is an inhomogeneous linear equation with constant coefficients, so its solution is the elementary quadrature of $e^{u\mathcal{M}}$:

$$\boxed{W(L) = L \varphi_1(L\mathcal{M})[D], \quad \varphi_1(z) \equiv \frac{e^z - 1}{z} = \sum_{k \geq 0} \frac{z^k}{(k+1)!}.} \quad (56)$$

The coordinate-free form matters because φ_1 is an *entire* function — the series on the right converges everywhere and $\varphi_1(0) = 1$ — so Eq. (56) needs no special case, no removable-pole treatment, and no genericity assumption on M_q . The mean-field transfer matrix of Sec. 13 is built from $\sinh(z)/z = \sum_{k \geq 0} z^{2k}/(2k+1)!$, also entire and also equal to 1 at the origin, so neither layer has a singularity at

a coalescing eigenvalue. An eigenbasis can therefore fail even when the finite- L matrix function is regular. This numerical fact does not make an exceptional point unphysical: eigenmode coalescence, non-normal amplification, and enhanced parameter sensitivity can remain observable properties of the drift.

18.3 Spectral resolution in trace–discriminant form

To see *which* exponent controls what, resolve Eq. (56) in the eigenbasis. If $M_q = S\Lambda S^{-1}$ with $\Lambda = \text{diag}(\mu_1, \mu_2)$, the eigenmatrices of $\mathcal{M}$ are $S_m S_n^\dagger$ and

$$\text{spec } \mathcal{M} = \{\mu_m + \mu_n^*\}, \quad (57)$$

so with $\tilde{D} = S^{-1}DS^{-\dagger}$,

$$W(L) = S \tilde{W}(L) S^\dagger, \quad \tilde{W}_{mn}(L) = \tilde{D}_{mn} L \varphi_1[(\mu_m + \mu_n^*)L] = \tilde{D}_{mn} \frac{e^{(\mu_m + \mu_n^*)L} - 1}{\mu_m + \mu_n^*}. \quad (58)$$

For the 2×2 two-mode block, define at each analysis frequency

$$s_q \equiv \frac{\text{tr } M_q}{2}, \quad q_q^2 \equiv \frac{(a_q - d_q)^2}{4} + b_q c_q, \quad \mu_\pm = s_q \pm q_q, \quad (59)$$

where a_q, b_q, c_q, d_q are the entries of $M_q(\Omega)$. Then Eq. (57) contains just three distinct values,

$$\mu_\pm + \mu_\pm^* = 2 \text{Re}(s_q \pm q_q), \quad \mu_\pm + \mu_\mp^* = 2 \text{Re } s_q \pm 2i \text{Im } q_q. \quad (60)$$

Within the fluctuation problem, $\text{Re}(s_q \pm q_q)$ set how fast each eigenchannel amplifies and accumulates diffusion, while $\text{Im } q_q$ sets the interference between channels. These equal the classical (s, q) only when $M_q(\Omega) = M_{\text{cl}}$. If $\text{Re}(\mu_m + \mu_n^*) < 0$ for every pair, $W(L)$ saturates as $L \rightarrow \infty$ at the algebraic Lyapunov solution $\tilde{W}_{mn} = -\tilde{D}_{mn}/(\mu_m + \mu_n^*)$; in an amplifier at least one exponent is positive and the noise grows with the gain, as it must.

18.4 Numerically stable evaluation

Equation (58) is the form to *reason* with, but it is not the form to compute with. It requires S to be invertible and well conditioned, and the parametric drift M_q is non-normal: as $q_q \rightarrow 0$ its two eigenvectors coalesce and S^{-1} diverges. The underlying $W(L)$ remains finite there, by Eq. (56); it is the coordinate system that fails, not the solution. Direct testing bears this out: the eigenbasis form loses all accuracy for a defective drift, while the coordinate-free form agrees with direct quadrature to 2×10^{-15} at the same point (Table 6).

Any stable evaluation of φ_1 therefore suffices. A convenient choice is the block exponential [7]: assemble

$$Z = \begin{pmatrix} -M_q & D \\ 0 & M_q^\dagger \end{pmatrix}, \quad e^{ZL} = \begin{pmatrix} F_{11} & F_{12} \\ 0 & F_{22} \end{pmatrix}, \quad (61)$$

from which

$$W(L) = F_{22}^\dagger F_{12}. \quad (62)$$

Equation (62) is Eq. (56) evaluated by a single 4×4 matrix exponential, with no eigenvector inversion anywhere. It is the recommended implementation.

18.5 Longitudinally varying drift

When M_q and D vary with z — through pump depletion, density gradients, or the Gaussian overlap profile — no single closed form exists. If M_k and D_k are the actual local coefficients, the segmented composition is exact within each piecewise-constant segment. Writing $T_k = e^{M_k dz}$ and $W_k = W(dz; M_k, D_k)$ for segment k , the covariance accumulates as

$$V^{(k+1)} = T_k V^{(k)} T_k^\dagger + W_k, \quad (63)$$

with each W_k evaluated by Eq. (62). This is the noise-carrying counterpart of the segmented mean-field propagation and introduces no approximation beyond piecewise-constant coefficients.

18.6 Static independent-vacuum completion fixture

For testing only, any constant drift may be given this particular local commutator completion. Using K from Eq. (50), let

$$K = U\Lambda_K U^\dagger, \quad B_{\text{vac}} = U|\Lambda_K|^{1/2}, \quad J_F = \text{diag}(\text{sgn } \Lambda_K), \quad D_{\text{vac}} = \frac{1}{2}B_{\text{vac}}B_{\text{vac}}^\dagger. \quad (64)$$

This eigenfactor closes each channel of K with one independent vacuum port and preserves the commutator by construction. The subscript “vac” denotes this chosen independent-vacuum algebraic fixture; it is not a universal lower bound over microscopic atomic reservoirs. In the special phase-insensitive amplifier setting, an analogous independent-vacuum construction gives the familiar Caves added-noise bound [5], but an arbitrary dissipative two-mode drift need not satisfy those assumptions. The fixture is not a derivation of atomic spontaneous-emission noise, and its symmetrized diffusion cannot be promoted to a physical bound without a microscopic model or data. When M_q and the diagnostic detector model are frequency independent, D_{vac} is also frequency independent, and feeding it through Eqs. (56) and (67) yields a flat S_- ; the machinery is exercised, but no bandwidth is predicted.

19 Present limitation of the frequency-dependent observable

The final in-cell observable must be a spectrum $S_-^{\text{out}}(\Omega_{\text{SA}})$, not a scalar. The algebraic path from $M_q(\Omega_{\text{SA}})$ and $D(\Omega_{\text{SA}})$ to that spectrum is now complete, but the microscopic $M_q(\Omega_{\text{SA}})$ and $D(\Omega_{\text{SA}})$ are not implemented. The static vacuum completion is paired here with a static detector model and yields only a flat diagnostic curve. It cannot reproduce:

- the squeezing bandwidth;
- low-frequency excess noise;
- the noise peak near 2 MHz from laser-source electronic noise;
- frequency-dependent atomic Langevin noise;
- slow-light and group-delay effects; or
- the frequency response of the detector.

Part VI

External detection model

20 External loss only after the cell

In the quadrature basis $\hat{\mathbf{R}} = (X_p, Y_p, X_c, Y_c)^T$, define

$$K_\eta = \text{diag}(\sqrt{\eta_p}, \sqrt{\eta_p}, \sqrt{\eta_c}, \sqrt{\eta_c}), \quad K_\ell = \text{diag}(\sqrt{1-\eta_p}, \sqrt{1-\eta_p}, \sqrt{1-\eta_c}, \sqrt{1-\eta_c}).$$

Only loss physically located after the cell is applied:

$$V_{R,\text{det}} = K_\eta V_{R,\text{out}} K_\eta^T + K_\ell V_{\text{vac}} K_\ell^T. \quad (65)$$

For symmetric efficiency,

$$V_{R,\text{det}} = \eta_{\text{ext}} V_{R,\text{out}} + (1 - \eta_{\text{ext}}) V_{\text{vac}}.$$

Atomic absorption and spontaneous emission remain distributed inside Eqs. (45)–(51). Cell optical-depth factors $\exp(-\text{OD}_{p,c,\text{seg}})$ are not applied a second time after propagation.

21 Electronic weighting and SQL normalization

The linearized photocurrent is

$$\delta\hat{i}_j(\Omega) \propto H_j(\Omega) \left[\alpha_j^* \delta\hat{a}_j(\Omega) + \alpha_j \delta\hat{a}_j^\dagger(-\Omega) \right], \quad \delta\hat{i}_- = \delta\hat{i}_p - w(\Omega) \delta\hat{i}_c. \quad (66)$$

The normalized spectrum is

$$S_-(\Omega) = \frac{\mathbf{m}^\dagger(\Omega) V_{R,\text{det}}(\Omega) \mathbf{m}(\Omega) + S_{\text{el}}(\Omega)}{S_{\text{SQL}}^{(\text{exp})}(\Omega)}. \quad (67)$$

Here $\mathbf{m}(\Omega)$ is the four-component measurement vector obtained from Eq. (66); it contains the detected carrier amplitudes, H_p, H_c , their phases, and the factor $-w$ on the conjugate quadratures. Equations (51), (52), and (62) supply the in-cell block $V_{\text{out}} = TV_{\text{in}}T^\dagger + W$ and hence $V_{R,\text{out}} = CV_{\Xi}C^\dagger$. Thus Eq. (67) is the single assembled observable: once the microscopic $M_q(\Omega_{\text{SA}})$, $D(\Omega_{\text{SA}})$, and measured detector functions are supplied, every remaining step is algebraic. With $V_{\text{vac}} = I/2$ and independent detector vacua, the corresponding ideal electronic standard-quantum-limit (SQL) term is proportional to

$$S_{\text{SQL}}(\Omega) \propto |H_p|^2 \eta_p |\alpha_p|^2 + w^2 |H_c|^2 \eta_c |\alpha_c|^2,$$

before the experiment-specific electronic calibration. The SQL reference must use the same detected powers, electronic transfer functions, weight w , resolution bandwidth, and video bandwidth. The report must state whether electronic dark noise is included or subtracted.

Table 4: Loss and noise ledger.

Mechanism	Physical location	Correct treatment
Atomic absorption and spontaneous emission	distributed in the vapor	Complex response in M_q plus associated B and D
Cell windows, filters, and free propagation	after the cell	External beam-splitter efficiency η_{path}
Detector quantum efficiency	detector face	External beam-splitter efficiency η_{QE}
Electronic imbalance and bandwidth	after photodetection	$H_p(\Omega)$, $H_c(\Omega)$, $w(\Omega)$, and calibrated $S_{\text{el}}(\Omega)$
Pump scattering into the detected mode	apparatus dependent	Measured technical-noise spectrum; not a universal atomic-loss formula

The ansatz

$$\mathcal{N}_{\text{ps}} = \kappa(1 - e^{-\text{OD}_{\text{pump}}})$$

is phenomenological. If used, it must be written $\mathcal{N}_{\text{ps}}(\Omega) = \kappa(1 - e^{-\text{OD}_{\text{pump}}})F_{\text{ps}}(\Omega)$ and calibrated to the experimental spectral normalization. It is not used as a universal Langevin noise law in this report.

Part VII Validation

22 Validation hierarchy

Agreement between this algebra and the production program is an algebraic and numerical consistency check, not independent physical validation. Independence is local rather than global: the continued-fraction code independently assembles the archived finite-Floquet blocks, while the literal

SI reference independently reconstructs Maxwell/ Q propagation from a shared reduced susceptibility. It does not independently reproduce the atomic Hamiltonian or dissipator. The separate pump reference checks its trace-constrained null vector against an SVD construction, its weak response against an independent dense/direct solve, and its rotating frame against the finite-Floquet state, while still sharing the same four-level atomic assembly. The levels below are therefore reported separately.

23 Verification of the analytic identities

The core closed forms used in the response, propagation, and noise layers were checked against independent direct computations before use. Table 6 records the residuals. The resolvent test uses the engine’s own 16×16 Liouvillian at the literature operating point; the Lyapunov tests use randomized non-normal drifts together with deliberately constructed defective cases.

The two **FAIL** rows are labeled “basis” because they are not failures of the solution. The same quantity computed from Eq. (56), at the same drift, is accurate to 2×10^{-15} ; only the eigenbasis resolution (58) degrades, and it degrades because S^{-1} does. That is the practical content of the remark in Sec. 18.2 that φ_1 is entire: there is nothing to regularize, only a coordinate system to avoid. A production implementation that diagonalizes M_q without a conditioning guard will silently return invalid results in exactly the near-degenerate regime where the two parametric eigenchannels are most strongly coupled.

24 Atomic pole spectrum at the operating point

Applying Eqs. (22)–(27) to the literature operating point ($\Delta/2\pi = +0.90$ GHz, $\delta/2\pi = -8$ MHz, $T = 121$ °C, $\Omega_{\text{pump}}/2\pi = 0.7076$ GHz) gives the pole structure in Table 7.

Power broadening widens the Raman pole fivefold. The slowest nonvanishing pole has half-width 0.5201 MHz, whereas the assumed ground-coherence rate is $\gamma_{gg}/2\pi = 0.1029$ MHz from Eq. (12). The pole is broadened by a factor 5.06. Any estimate that identifies the Raman feature width with γ_{gg} is therefore low by a factor of 5.06 at this pump power. This is the quantitative content of the statement in Sec. 10 that $-\text{Re } \lambda_m$ is not a bare rate.

A narrow Raman feature remains in the one-dimensional axial average. The averaged response is not itself a finite pole sum. Rather, on the one-dimensional axial grid used for Table 7, the per-velocity Raman-pole centers remain at zero while their half-widths vary over 0.220–1.494 MHz: the two-photon combination cancels the common axial Doppler shift. This does *not* include the transverse term of Eq. (17), which supplies the 1.38 MHz rms center spread in the full two-dimensional geometry. The separate reference in Table 3 includes that term; the pole table itself remains one-dimensional. The Raman feature is therefore broadened but not smeared over the 247 MHz optical scale.

The separate scales do not determine a squeezing bandwidth. The per-velocity Raman poles supply Lorentzian half-widths of 0.22–1.49 MHz, while the non-collinear term supplies a Gaussian rms center spread of 1.38 MHz. These quantities have different definitions: a Lorentzian has no finite variance, and the maximum half-width of one velocity class is not the half-width of the averaged line. They therefore cannot be combined by a root-sum-of-squares rule. The measured 3.5 MHz bandwidth belongs to $S_-(\Omega_{\text{SA}})$ after propagation and detection, not to a bare atomic pole. The two-dimensional mean-field response is now available as a separate reference, but a quantitative comparison still requires frequency-dependent $M_q(\Omega_{\text{SA}})$ and $D(\Omega_{\text{SA}})$ plus detector transfer functions. Until those are supplied, the pole table is a scale diagnosis and no numerical bandwidth gap is claimed.

25 Floquet-truncation convergence

Table 8 repeats the archived-model calculation at the legacy numerical optimum $\Delta/(2\pi) = -1.50$ GHz, $T = 110^\circ\text{C}$ and approximately $\delta/(2\pi) = -280$ MHz. It deliberately retains the archived dressed-wave-vector phase convention only to isolate Floquet convergence; because that convention double counts dispersion, these numbers are not the Option-A physical prediction. To prevent their legacy columns from being mistaken for canonical gains, the next two tables use $g_{c,\text{hist}} \equiv |T_{21}|^2$ and $\Delta_{\text{hist}} \equiv G_p - g_{c,\text{hist}}$. Only their relative truncation behavior is used.

The relative changes from $N_F = 1$ to 2 are

$$\frac{|G_p^{(2)} - G_p^{(1)}|}{G_p^{(2)}} = 16.80\%, \quad \frac{|g_{c,\text{hist}}^{(2)} - g_{c,\text{hist}}^{(1)}|}{g_{c,\text{hist}}^{(2)}} = 12.97\%.$$

The historical coefficient gap even changes sign. Thus the $N_F = 1$ result is *qualitative and unconvverged*. The changes from $N_F = 2$ to $N_F = 3$ are below $4 \times 10^{-7}\%$ at the tested point. Direct block solves and the independent assembly agree to 5.4×10^{-13} or better. Convergence at one operating point does not prove global convergence; this test must be repeated for every reported scan. The production path now performs that full-scan $N_F/N_F - 1$ comparison and returns **UNCONVERGED** rather than silently promoting a failed scan.

For the propagation-only Option-A correction at the fixed literature point, the inherited approximate atomic response gives the results in Table 10. Here $N_F = 1 \rightarrow 2$ changes G_p by 0.4528% and G_c by 0.4906%; $N_F = 2$ and 3 agree at the displayed precision. This establishes Floquet convergence of that fixed finite-seed propagation benchmark. It does not use the pump-only reference; the comparison with that reference is reported separately below. The multidimensional-velocity approximation remains untouched. The displayed gains assume equal collected mode areas, but the conversion is explicit: $T_{\text{can}} = Q^{-1}T_E Q$, $G_c = (\omega_c/\omega_p)|T_{\text{can},21}|^2$, and $\mathcal{G}_{\text{flux}} = G_p - |T_{\text{can},21}|^2$.

The inherited weak-response implementation also retains a finite reference probe-seed. Its numerical linearity check is

Reference probe-seed power	G_p	G_c	$\mathcal{G}_{\text{flux}}$
2 μW	403.752756	406.102206	-2.342895
4 μW	403.680319	406.019313	-2.332440
8 μW	403.535497	405.853591	-2.311543

Between 2 and 8 μW , G_p changes by -0.05381% , G_c by -0.06122% , and the signed flux gap changes by -1.3382% relative to its 2 μW value. The absolute gains are weakly seed dependent, while the noncanonical flux gap is relatively sensitive. This supports numerical linearity with respect to the reference probe-seed at this fixed point only; by itself it does not validate the underlying pump-state approximation.

25.1 Pump-only reference and the finite-seed limit

The independent reference closes that specific numerical gap for the standard minus Raman branch without replacing the production solver. On the complete configured one-dimensional Doppler grid, four analysis frequencies $\Omega_{\text{SA}}/2\pi = (0, 0.1, 1, 4)$ MHz and five two-photon detunings $\delta/2\pi = (-30, -8, 0, 12, 40)$ MHz give the diagnostics in Table 11.

The finite-reference susceptibility was then compared with the infinitesimal derivative after Doppler averaging all four complex response components over the same five detunings. Table 12 reports the worst component error normalized by the maximum pump-reference response.

The errors decrease monotonically, and a log-log fit to the last three rows has Rabi-amplitude slope 1.749. The finite range therefore supports convergence to the infinitesimal reference but does *not* establish an exact quadratic law. At the smallest amplitude, $N_F = 1$ gives a 4.485×10^{-3} worst error, whereas $N_F = 2$ and 3 both give 9.138×10^{-6} ; seed linearization and Floquet truncation must

therefore be checked separately. The direct resolvent and diagonalizable pole/residue sum agree to 2.45×10^{-24} s or better over the declared analysis band, with eigenvector condition number 3.39. The projected zero-relative-frequency fixture has normalized residual 1.79×10^{-15} .

26 Physicality tests do not detect the Floquet truncation error

Since ρ_0 is the time average of a physical $\rho(t)$, it must be Hermitian and positive semidefinite, with unit trace. Table 13 checks those necessary conditions for the $N_F = 1$ truncation at both operating points, after the convergence failure above has already been quantified.

Every test passes, including at the archived point where Table 8 shows a 16.8% error in G_p and a sign reversal of the historical transfer-coefficient gap. State positivity is therefore insensitive to this truncation error: a valid density matrix can be quantitatively wrong. The exact-GKSL half-plane condition of Sec. 10.4 is an assembly invariant for the same reason, so neither check substitutes for explicit N_F convergence.

Table 5: Validation hierarchy. No combined error column is used.

Level	Test performed	Status
Algebraic consistency	Closed-form matrix exponential versus direct exponential; dimensional and frequency audits	PASS
Analytic-identity verification	Spectral resolvent expansion, Lyapunov closed form, block-exponential form, and the cubic characteristic polynomial against direct numerics	PASS (Table 6)
Numerical implementation consistency	Continued-fraction versus direct finite-block assembly for the archived reference, plus production Maxwell/ Q versus a literal SI propagation reference	PASS within shared atomic assumptions
Pump-only weak-response reference	Minus-branch static-frame/Floquet gauge parity, trace-one state, trace-zero finite-frequency and projected-DC response, finite-seed limit, and direct/pole-residue parity	PASS numerically within the four-level, one-dimensional-velocity model; separate slow reference, plus branch unsupported
Two-dimensional Raman-Doppler reference	Independent (v_z, v_x) Maxwell quadrature, lab-frequency invariance, analytic rms, collinear/equal- k limits, grid refinement, and cutoff refinement	PASS for the separate slow minus-branch reference; production remains one-dimensional
Truncation convergence	$N_F = 1, 2, 3$ direct/continued/kernel parity and scan-level adjacent-order checks	Production $N_F = 3/2$ gate implemented; historical $N_F = 1$
Independent microscopic calculation	Multilevel Zeeman-resolved Liouvillian and atomic diffusion matrix	FAIL
Experimental gain comparison	Reduced model evaluated at the documented ⁸⁵ Rb operating point	not yet available
Experimental bandwidth comparison	Full $S_-(\Omega_{SA})$ versus the measured 3.5 MHz squeezing bandwidth	performed; substantial discrepancy
Experimental squeezing-spectrum comparison	Full $S_-(\Omega_{SA})$ using microscopic $D(\Omega_{SA})$	two-dimensional mean-field reference complete only for the declared minus branch, angle, and quadrature; microscopic diffusion and detector response pending
		not yet available

Table 6: Verification of the analytic results against direct numerics. Relative errors are Frobenius norms unless stated otherwise.

Identity	Reference computation	Relative error	Verdict
Spectral resolvent expansion, Eq. (22)	Direct linear solve on the 16×16 engine Liouvillian, $\Omega_{\text{SA}}/2\pi = 0.1\text{--}4$ MHz	1.0×10^{-12}	PASS
Vanishing of the steady-state residue, Eq. (26)	$\langle l_0 \mathcal{L}$ and $\langle l_0 \mathcal{V}_k \rho_{\text{ss}} \rangle$ on the engine Liouvillian	exactly zero in floating-point arithmetic	PASS
Coordinate-free solution, Eq. (56)	Simpson quadrature of Eq. (53); generic, defective, and coalescing drifts	2.2×10^{-15} (worst)	PASS
Eigenbasis resolution, Eq. (58)	Same quadrature, randomized non-normal M_q , PSD D	5.7×10^{-15}	PASS
Hermiticity and positivity of W	Same, invariants of Eq. (55)	1.0×10^{-16} ; $\min \text{eig } W > 0$	PASS
Eigenbasis resolution at a defective drift	Same quadrature, exact Jordan block	loses all accuracy	FAIL (basis)
Eigenbasis resolution near an exceptional point	Same quadrature, coalescing eigenvalues	1.7×10^{-1}	FAIL (basis)
Lyapunov superoperator spectrum, Eq. (57)	$\text{spec } \mathcal{M}$ versus $\{\mu_m + \mu_n^*\}$, 200 random cases up to 4×4	1.5×10^{-13}	PASS
Block-exponential evaluation, Eq. (62)	Same quadrature, generic and defective drifts	8.5×10^{-15} (worst)	PASS
Cayley–Hamilton collapse, Eq. (38)	$\ N^2 - q^2 I\ $ and the resulting $e^{M_{\text{cl}} L}$ versus a direct exponential, including $q = 0$	1.5×10^{-14}	PASS
Characteristic polynomial, Eq. (76)	Direct eigenvalues of Eq. (75)	1.7×10^{-15}	PASS
Undamped limit, Eq. (77)	Roots versus $\Omega' = \sqrt{\Omega^2 + \Delta^2}$	exact to machine precision	PASS

Table 7: Slowest Liouvillian poles at the literature operating point, zero-velocity class. Half-width is $-\text{Re } \lambda_m/2\pi$ and center is $-\text{Im } \lambda_m/2\pi$. The Doppler column is the range of the half-width across the one-dimensional axial velocity grid.

Half-width [MHz]	Center [MHz]	Doppler range of half-width [MHz]	Assignment
0 (exact)	0	—	Steady state; null eigenvalue of Eq. (9)
0.5201	0	0.220–1.494	Ground-state Raman coherence, power broadened
0.5894	± 190.05	0.311–1.494	Excited-state hyperfine coherence
3.3024	± 1023.57	0.311–1.549	Optical/dressed coherence

Table 8: Floquet convergence of the archived reduced model at the fixed point $\delta/(2\pi) = -280$ MHz. Sectors $n = -N_F, \dots, N_F$ are retained.

N_F	G_p	$g_{c,\text{hist}}$	Δ_{hist}	$\arg \chi_{pc}$	$\arg \chi_{cp}$
1	22.5913	21.9687	+0.622609	178.142°	−179.333°
2	19.3415	19.4472	−0.105684	178.173°	−179.366°
3	19.3415	19.4472	−0.105684	178.173°	−179.366°

Table 9: Separate minimization of the gain-only objective. All observables in a row are evaluated at that row's δ^* . The scan spacing is 5 MHz.

N_F	$\delta^*/(2\pi)$ [MHz]	G_p	$g_{c,\text{hist}}$	Δ_{hist}	$\arg \chi_{pc}$	$\arg \chi_{cp}$
1	−280	22.5913	21.9687	0.622609	178.142°	−179.333°
2	−275	8.97871	8.46840	0.510314	178.402°	−179.582°
3	−275	8.97871	8.46840	0.510314	178.402°	−179.582°

Table 10: Floquet convergence at the fixed literature point $\Delta/(2\pi) = +0.90$ GHz, $\delta/(2\pi) = -8$ MHz, and $T = 121$ °C. This is not an optimization; it is the production-like finite-seed path, not the separate pump-only reference.

N_F	G_p	G_c	$g_c^{(\Phi)}$	$\mathcal{G}_{\text{flux}}$	$\arg \chi_{pc}$	$\arg \chi_{cp}$
1	401.70833	403.86243	403.85591	−2.147574	−179.886152°	179.814904°
2	403.53550	405.85359	405.84704	−2.311543	−179.885623°	179.814496°
3	403.53550	405.85359	405.84704	−2.311543	−179.885623°	179.814496°

Table 11: Executable pump-only state and trace-zero weak-response diagnostics.

Check	Maximum error or margin
Normalized pump-state equation residual	4.00×10^{-17}
Pump-state trace error	2.89×10^{-15}
Minimum pump-state eigenvalue	2.371×10^{-3}
Normalized weak-response equation residual, full 4×5 scan	7.32×10^{-17}
Weak-response trace error, full 4×5 scan	6.48×10^{-25}
Static-frame versus $N_F = 3$ pump-only Floquet state	8.37×10^{-15}
Projected zero-relative-frequency response residual	1.79×10^{-15}
Direct resolvent versus pole/residue sum, 0–4 MHz analysis band	2.45×10^{-24} s
Pole eigenvector condition number	3.39

Table 12: Finite-seed $N_F = 3$ approach to the pump-only infinitesimal reference.

Rabi-amplitude fraction	Seed power (μ W)	Worst normalized error
1.00	8.0000	2.503×10^{-3}
0.30	0.7200	5.093×10^{-4}
0.10	0.0800	9.254×10^{-5}
0.03	0.0072	9.138×10^{-6}

Table 13: Physicality of the $N_F = 1$ truncated Floquet state. Minima are taken over the velocity grid and, for $\rho(t)$, over one beat period.

Quantity	Archived point	Literature point	Requirement
$\max \ \rho_0 - \rho_0^\dagger\ _\infty$	7.4×10^{-14}	3.3×10^{-14}	0
$\max \text{Tr } \rho_0 - 1 $	1.1×10^{-15}	5.6×10^{-15}	0
$\min \text{eig } \rho_0$	$+7.90 \times 10^{-3}$	$+2.42 \times 10^{-3}$	≥ 0
$\min_t \text{eig } \rho(t)$	$+5.63 \times 10^{-3}$	$+2.28 \times 10^{-3}$	≥ 0

27 Velocity-integration convergence

The following tests retain the same historical propagation columns and use $N_F = 3$ with the one-dimensional Doppler integral. They validate only that quadrature, not the propagation convention or non-collinear geometry. At a fixed $5\sigma_v$ cutoff, velocity-step convergence is excellent:

Table 14: One-dimensional velocity-step convergence at fixed $|v_z| \leq 5\sigma_v$.

Δv [m/s]	Classes	G_p	$g_{c,\text{hist}}$	Relative G_p error
10.00	195	18.936590	19.024328	$1.56 \times 10^{-3}\%$
5.00	389	18.936295	19.024021	$2.66 \times 10^{-7}\%$
2.50	777	18.936295	19.024021	$9.58 \times 10^{-8}\%$
1.25	1551	18.936295	19.024021	reference

The cutoff, however, is important:

Table 15: One-dimensional cutoff convergence at $\Delta v = 2.5$ m/s.

Cutoff	G_p	$g_{c,\text{hist}}$	Δ_{hist}	Relative G_p error
$3.0\sigma_v$	19.36195	19.46857	-0.10662	2.248%
$3.5\sigma_v$	19.11527	19.21041	-0.09514	0.945%
$4.0\sigma_v$	18.98214	19.07172	-0.08957	0.242%
$4.5\sigma_v$	18.93642	19.02414	-0.08773	$6.4 \times 10^{-4}\%$
$5.0\sigma_v$	18.93629	19.02402	-0.08773	reference

For the experimental phase-matching angle, the separate slow reference now integrates Eq. (18) and passes the grid/cutoff criteria in Table 3. Its angular rms width $\sigma_{\delta,\theta}/(2\pi) = 1.38017$ MHz is not negligible relative to $\gamma_{gg}/(2\pi) = 0.103$ MHz and, as Table 7 shows, is comparable to the power-broadened Raman pole width itself. This result does not certify the one-dimensional finite-seed production scan.

28 Commutator-preservation test

At the literature operating point, assume for this diagnostic that the collected probe and conjugate have equal effective Gaussian mode areas, $A_p = A_c = 1.7105972 \times 10^{-7} \text{ m}^2$. Their distinct optical frequencies are retained in Q , rather than replaced by $Q \propto I$. The resulting photon-flux-normalized static drift in the symmetric mismatch gauge is

$$M_q = \begin{pmatrix} -1.795444 + 8.518638i & -0.593347 + 297.229796i \\ 0.962238 - 297.200894i & 0.508740 + 60.465455i \end{pmatrix} \text{ m}^{-1}. \quad (68)$$

If the actual conjugate collected-mode area differs from the equal-area audit assumption, $Q^{-1}M_EQ$ must be recomputed before applying the canonical metric J . The normalization operation itself is now implemented; the equal-area apparatus input remains conditional. For the numerical test only, factor the Hermitian matrix $K = -(M_qJ + JM_q^\dagger)$ as $K = BJ_FB^\dagger$. Here its eigenvalues are 0.285202 m^{-1} and 4.323166 m^{-1} . Both are positive, so by Sec. 16 the displayed independent-vacuum algebraic completion with $J_F = I$ exists: at this point the drift requires loss-type reservoir channels. This construction proves that the drift can be embedded in a canonical channel; it is not a microscopic derivation of atomic noise.

Table 16: Canonical-commutator audit.

Test	Residual	Result
Option-A drift with reservoir omitted	$\max TJT^\dagger - J = 3.388$	FAIL
Local identity with the explicit Q transform	$\max M_qJ + JM_q^\dagger + BJ_FB^\dagger = 4.44 \times 10^{-16} \text{ m}^{-1}$	PASS conditional
Integrated identity through the cell, explicit Q	$\ TJT^\dagger + \int T_z BJ_FB^\dagger T_z^\dagger dz - J\ _F / \ J\ _F = 3.71 \times 10^{-13}$	PASS conditional
Microscopic $M_q(\Omega)$, $B(\Omega)$, and $D(\Omega)$ over 0.1–4 MHz	Not implemented in the current four-level code	not yet available

Because canonical commutator preservation constrains only the antisymmetric reservoir correlations, it fixes neither the symmetrized diffusion matrix $D(\Omega)$, nor its temperature, nor excess spontaneous-emission noise; the vacuum-fixtured values below therefore serve as illustrative diagnostics for one chosen vacuum completion, not as bounds or Langevin-corrected physical predictions.

29 Legacy gain-only diagnostic and its failure domain

The canonical ideal-amplifier limit must be separated from the historical gain-only scan score. Define the conjugate photon-flux coefficient

$$g_c^{(\Phi)} \equiv |T_{\text{cl},21}|^2 = \frac{\omega_p}{\omega_c} G_c. \quad (69)$$

For the two-mode map

$$\hat{a}_p^{\text{out}} = \mu \hat{a}_p^{\text{in}} + \nu \hat{a}_c^{\text{in}\dagger}, \quad |\mu|^2 = G_p, \quad |\nu|^2 = g_c^{(\Phi)},$$

the output commutator is $G_p - g_c^{(\Phi)} = \mathcal{G}_{\text{flux}}$. Only when $\mathcal{G}_{\text{flux}} = 1$ is this map canonical without additional reservoir operators. In that lossless limit the photon-number difference is conserved, and a bright coherent probe seed with a vacuum conjugate input gives

$$S_{\text{ideal}} = \frac{1}{G_p + g_c^{(\Phi)}} = \frac{1}{2G_p - 1}. \quad (70)$$

Symmetric external efficiency η then gives

$$S(\eta) = 1 - \eta + \eta S_{\text{ideal}}. \quad (71)$$

For an ideal canonical amplifier with unequal efficiencies or a weighted difference, set $A = G_p$, $B = g_c^{(\Phi)}$, and $A - B = 1$. In units of the incident seed photon flux, the exact bright-seed variance and SQL are

$$\begin{aligned} \mathcal{V}_w &= \eta_p^2 A(A + B) + \eta_p(1 - \eta_p)A \\ &\quad + w^2 [\eta_c^2 B(A + B) + \eta_c(1 - \eta_c)B] - 4w\eta_p\eta_c AB, \end{aligned} \quad (72)$$

$$S_w = \frac{\mathcal{V}_w}{\eta_p A + w^2 \eta_c B}. \quad (73)$$

Here $w = 1$ is a raw difference and $w = \eta_p A / (\eta_c B)$ balances the detected DC levels. For unequal losses the weight that minimizes normalized noise is covariance dependent; it cannot in general be inferred from detected powers alone. Equation (73) reduces to Eq. (71) when $w = 1$ and $\eta_p = \eta_c$.

The archived optimizer instead evaluated the numerical score

$$S_{\text{legacy}} = \text{clip}_{[0,1]} \frac{(G_p - g_c^{(\Phi)})^2}{G_p + g_c^{(\Phi)}}. \quad (74)$$

This is a definition retained only to reproduce historical scans. It is not a covariance and its clipping does not prevent passive points from giving $S_{\text{legacy}} < 1$; for example $(G_p, g_c^{(\Phi)}) = (0.2, 0.1)$ gives $S_{\text{legacy}} = 0.033$. For the dissipative cell, $\mathcal{G}_{\text{flux}} \neq 1$ requires the reservoir channels of Secs. 16 and 18. Unequal detector weights must then be applied to the full covariance through Eq. (67), not reconstructed from the two mean gains.

30 Model point and experimental comparison

This benchmark applies the passive-limit-tested Option-A propagation signs and $N_F = 3$ to the inherited finite-reference-field four-level Floquet response. It is not an output of the pump-null-space and frequency-resolvent formulation of Parts II–III. It uses a $5\sigma_v$ one-dimensional velocity range and the documented point

$$\frac{\Delta}{2\pi} = +0.90 \text{ GHz}, \quad \frac{\delta}{2\pi} = -8 \text{ MHz}, \quad T = 121 \text{ }^\circ\text{C}, \quad L = 12.5 \text{ mm},$$

with 600 mW pump power, 8 μ W probe-seed power, $w_{\text{pump}} = 530 \text{ } \mu\text{m}$, $w_p = 330 \text{ } \mu\text{m}$, and $\theta = 0.32^\circ$. The radii are $1/e^2$ intensity radii. The calculation omits the transverse two-photon Doppler average and therefore remains a reduced-model benchmark. In the same symmetric mismatch gauge used for Eq. (68), the transfer is

$$T_E = \begin{pmatrix} 18.839694 + 6.971473i & -8.603522 + 18.218019i \\ 8.625460 - 18.205907i & 17.483771 + 10.224863i \end{pmatrix}.$$

For equal collected mode areas, this gives

$$G_p = 403.53550, \quad G_c = 405.85359, \quad g_c^{(\Phi)} = 405.84704, \quad \mathcal{G}_{\text{flux}} = -2.31154.$$

Here T_E is the electric-field transfer; applying the explicitly constructed $Q^{-1}T_EQ$ gives $g_c^{(\Phi)} = (\omega_p/\omega_c)G_c$ with $\omega_c/\omega_p = 1.00001614$. The former (32.56066, 31.23207) result used the opposite susceptibility-propagation sign and fails the passive Beer–Lambert test. The (5.61434, 4.66816) result used the passive susceptibility sign but retained both the duplicated $p_F/[2(2I+1)]$ population factor and the wrong symmetric-frame mismatch sign. The intermediate (107.64682, 107.36704) result removed the population duplicate but still used that wrong mismatch sign. The later (1054.30596, 1061.88534) result corrected those conventions but represented the inherited transit floor as coherence-only damping. The current result adds the trace-preserving thermal reload channel. None of the earlier numbers is retained after the combined carrier, mismatch, normalization, and transit audit. The inherited residual $\ell_s = 0.74$ has not been refitted to compensate.

Table 17: Separation of mean-field gain and quantum-noise layers at the literature operating point, using explicit Q with an equal collected-area assumption. The symmetric external efficiency used in the diagnostic columns is $\eta_{\text{ext}} = 0.8694$.

Layer	G_p	G_c	S_-^{out}	S_-^{det}	Interpretation
Semiclassical reduced model	403.535	405.854	—	—	Mean-field gain only
Ideal Bogoliubov amplifier matched to G_p	403.535	402.542	0.001241 (−29.06 dB)	0.13168 (−8.80 dB)	Lossless limiting case, not the dissipative medium
Static independent-vacuum completion fixture	403.535	405.854	0.02414 (−16.17 dB)	0.15158 (−8.19 dB)	Canonical diagnostic with $w = 1$; D is not microscopic
Static independent-vacuum completion fixture, DC balanced	403.535	405.854	0.005233 (−22.81 dB)	—	Same diagnostic with $w \simeq 0.99429$; detector rematching not evaluated
Microscopic quantum-Langevin model	403.535	405.854	not yet available	not yet available	Required physical calculation
Experiment [1]	111/8 = 13.875	109/8 = 13.625	—	0.166 (−7.8 dB)	Raw 8 μ W input and 111/109 μ W outputs; both power ratios are independently available

The static completion and detector model are flat by construction, so repeating them at several analysis frequencies adds no information. The measured features instead show what is missing: technical excess noise below a few hundred kHz, -8.7 dB from the IDS/SQL slope ratio at 150 kHz, a laser-source noise peak near 2 MHz, and loss of squeezing beyond roughly 3.5 MHz [1]. None can be reconstructed from two gains, so the table reports no physical $S_-(\Omega_{\text{SA}})$ prediction.

Table 18: Model–experiment comparison after correcting propagation signs and removing duplicate manifold-population weighting.

Observable	Reduced model	Experiment or benchmark	Discrepancy
G_p	403.535	13.875 from raw powers	+2808.4%
G_c	405.854	13.625 from raw powers	+2878.7%
$\delta^*/(2\pi)$	-275 MHz in the archived reduced scan	-8 MHz measured operating point	-267 MHz; different objective and incomplete geometry make this nonpredictive
$\Delta^*/(2\pi)$	-1.50 GHz in the archived reduced scan	$+0.90$ GHz measured operating point	-2.40 GHz
Maximum squeezing	no physical prediction	-7.8 dB	Microscopic diffusion and detector response missing
Squeezing bandwidth	no physical prediction; separate Lorentzian pole and Gaussian Doppler scales are tabulated	approximately 3.5 MHz	Two-dimensional mean-field reference complete; microscopic $S_-(\Omega_{\text{SA}})$ comparison pending (Sec. 24)

The point $(\Delta/2\pi, \delta/2\pi, T) = (-1.50 \text{ GHz}, -280 \text{ MHz}, 110^\circ\text{C})$ is classified only as a *numerical optimum under the archived reduced model*. It is not an experimentally established optimum. Likewise, the numerical proximity of the chosen-vacuum algebraic fixture to the measured -7.8 dB is not validation: the former contains no microscopic atomic diffusion or detector spectrum.

31 Required limiting cases

Table 19: Analytic limiting-case checks for the layered model.

Limit	Required result	Status
No atom-field coupling	$\chi_{ij} \rightarrow 0$ implies $G_p \rightarrow 1$, $G_c \rightarrow 0$, $D \rightarrow 0$, and $S_- \rightarrow 1$	PASS analytically
Ideal lossless parametric amplifier	$M_q J + J M_q^\dagger = 0$, $B = D = 0$, and $\mathcal{G}_{\text{flux}} = 1$	PASS analytically
Bright probe-seeded ideal amplifier	$S_{\text{ideal}} = 1/[G_p + g_c^{(\Phi)}]$, reducing to $1/(2G - 1)$ when $\mathcal{G}_{\text{flux}} = 1$	PASS analytically
Symmetric external efficiency	$S_-^{\text{det}} = 1 - \eta + \eta S_-^{\text{out}}$	PASS analytically
Zero phase-matching angle	$\theta \rightarrow 0$ removes $k\theta\sigma_v$ from the two-photon Doppler width	PASS analytically
Zero dissipation	The Langevin contribution vanishes and $TJT^\dagger = J$	PASS as a formal limit
Zero cell length	$W(0) = 0$ from Eq. (55), so $V_{\text{out}} = V_{\text{in}}$	PASS analytically and numerically
Undamped two-level drive	Poles at $0, \pm i\sqrt{\Omega^2 + \Delta^2}$ from Eq. (76)	PASS numerically
Vanishing drive in the pole sum	$\mathcal{V}_k \rightarrow 0$ makes every residue $A_m^{(jk)}$ vanish, so $\bar{\chi} \rightarrow 0$	PASS analytically

32 Conclusions and validation status

The definitive status is the hierarchy in Table 5. The analytic core — the trace-zero resolvent, the diagonalizable-case pole sum with its Jordan/contour qualification, constant-matrix transfer, commutator identities, Lyapunov solution, and external detection map — is closed, with the identities in Table 6 verified to 10^{-12} or better. The production finite-Floquet layer now adds an explicit full-scan adjacent-order gate; the separate velocity audits establish numerical convergence only for their stated reduced problems. State positivity and the exact-GKSL half-plane condition remain necessary invariants, not substitutes for convergence or physical validation. The separate minus-branch pump reference additionally closes the executable trace-one state, trace-zero response, finite-seed-limit, and direct pole-sum checks at the errors reported in Table 11. It remains a slow reference and does not change the finite-seed production default; the inherited plus-branch reference remains unsupported. The separate tensor-Maxwell wrapper additionally closes the declared two-dimensional grid and cutoff checks for a single crossing angle, with the lab beat held fixed.

At the literature operating point, correcting the Maxwell propagation signs changes the reduced-model gains to those of Table 18; their large disagreement with the independently reported output powers precludes quantitative gain validation. The independent-vacuum completion is retained solely as a commutator-preserving algebra test. No gain-only number in this report is a squeezing prediction, and the separate Raman-pole and angular-Doppler scales do not determine a squeezing bandwidth.

A physical $S_-(\Omega_{\text{SA}})$ still requires several coupled inputs: promotion of the certified minus-branch atomic response into a production frequency-dependent drift (and a valid treatment of any claimed plus branch), microscopic reservoir coupling and diffusion, promotion of the certified two-dimensional response into the finite-seed/segmented production path plus a measured beam angular distribution, collected-mode normalization, and measured detector transfer and electronic-noise functions. Until those are implemented together, the document is a semianalytic mean-field reconstruction and quantum-model specification, not a validated squeezing calculation.

A Analytically solvable limit: the damped two-level rotor

The full four-level Liouvillian is 16×16 and its characteristic polynomial has degree 16, so its roots are computed numerically. A reduced two-level case supplies the physical labels used in Sec. 10. With longitudinal and transverse relaxation times T_1 and T_2 , detuning Δ , and Rabi frequency Ω , the Bloch generator in the (u, v, w) basis is

$$M_B = \begin{pmatrix} -1/T_2 & -\Delta & 0 \\ \Delta & -1/T_2 & \Omega \\ 0 & -\Omega & -1/T_1 \end{pmatrix}, \quad (75)$$

with the factored characteristic polynomial

$$\left(\lambda + \frac{1}{T_2} \right) \left[\left(\lambda + \frac{1}{T_2} \right) \left(\lambda + \frac{1}{T_1} \right) + \Omega^2 \right] + \Delta^2 \left(\lambda + \frac{1}{T_1} \right) = 0. \quad (76)$$

Its informative limits are immediate:

- **Undriven**, $\Omega = 0$: the population pole is $-1/T_1$ and the free-precession pair is $-1/T_2 \pm i\Delta$.
- **Resonant**, $\Delta = 0$: the u quadrature has pole $-1/T_2$; the remaining quadratic gives a damped-Rabi pair only when $|\Omega| > |T_1^{-1} - T_2^{-1}|/2$, and otherwise gives two additional real overdamped branches.
- **Undamped**, $T_1, T_2 \rightarrow \infty$:

$$\lambda_0 = 0, \quad \lambda_{\pm} = \pm i\Omega', \quad \Omega' = \sqrt{\Omega^2 + \Delta^2}. \quad (77)$$

Thus the roots comprise one population-recovery branch and either a complex-conjugate damped-Rabi pair or two additional real overdamped branches. This is the structure named by Eq. (27) in the multilevel case. Equation (76) follows Torrey’s classical transient-nutation analysis [6].

B Dimensional-consistency audit

Table 20: Dimensional audit in SI units.

Quantity	Dimension	Check
$\Delta, \delta, \Gamma, \Omega_{\text{SA}}, \Delta_e, \lambda_m$	s^{-1}	Angular frequencies and Liouvillian poles
$\bar{\chi}_{ij}; Nd^2/(\epsilon_0 \hbar)$	$\text{s}; \text{s}^{-1}$	Their product is dimensionless χ_{ij}
M_{cl}, M_q, D, K	m^{-1}	Spatial drift, diffusion, and commutator defect per unit length
$\mu_{\pm}$	m^{-1}	Eigenvalues of a spatial generator
$W(L)$	1	$\int D dz$ with dimensionless covariance
$V(x; \sigma, \gamma)$	s	$-\pi V$ is an inverse angular frequency
G_p, G_c, S_-	1	Power gains and normalized noise spectrum

C Assumptions and validity domains

Table 21: Assumption and validity-domain ledger.

Layer	Assumption	Validity domain and failure mode
Atomic basis	Four hyperfine manifolds with Zeeman structure averaged into effective strengths	Hyperfine-scale qualitative trends; fails for polarization, optical pumping, dark states, and magnetic response
Pump-only weak-response reference	Undepleted prescribed pump, static four-level Markovian Liouvillian, trace-one null space, and trace-zero Nambu resolvent	Executable slow reference for the minus Raman branch only; not used by the current gain tables, plus branch unsupported, and controlled only under the stated phenomenological decay channels; a separate 2-D wrapper is available
Current gain-state path	Trace-one finite-seed density matrix; production $N_F \geq 3$ with an $N_F - 1$ full-scan gate, audit $N_F = 1, 2, 3$	Not a pump-only state; convergence and finite-seed back-action must be reported separately
Historical rate/Sylvester model	Populations solved without coherence feedback	Weak saturation or rapid dephasing; not an exact strong-pump steady state
Weak fields	Probe and conjugate enter to first order	Valid below the onset of pump depletion, gain saturation, and nonlinear probe back-action

Continued on next page

Layer	Assumption	Validity domain and failure mode
Pole expansion	$\mathcal{L}$ is diagonalizable	Exact where it holds; defective Liouvillian eigenvalues require a Jordan or contour form with higher-order poles, and eigen-residues become ill conditioned near defective or nearly defective points
Lorentzian reading of poles	Residues vary slowly over the analysis band	Overlapping poles with comparable residues interfere and do not decompose into independent visible peaks
Floquet expansion	Finite N_F	Must be increased until observables converge; $N_F = 1$ fails at the archived point, and no positivity test detects the failure
Velocity average	Maxwell distribution, independent Cartesian components	The separate minus-branch reference resolves (v_z, v_x) for one declared angle; the finite-seed production average remains one-dimensional and omits beam divergence
Classical propagation	z -independent coefficients and fixed density, temperature, and pump	Closed-form exponential; fails with longitudinal pump attenuation, density gradients, or depletion
Lyapunov solution	Constant M_q , D over the integration interval	No condition on M_q ; use the segmented recursion for varying coefficients. Only the eigenbasis <i>resolution</i> additionally needs a well-conditioned S
Phase convention	Diagonal complex susceptibility plus vacuum/geometric mismatch only	Avoids double counting of $\text{Re } \chi$
Quantum propagation	Linear Gaussian Langevin channel	Requires a microscopic reservoir matrix and appropriate Markov approximation
Static vacuum completion	Minimal-port independent-vacuum factorization of the drift	Commutator fixture only; neither a universal Caves-bound claim nor an atomic-noise prediction, and frequency independent by construction
Detection	External path and detector losses act after the cell	Valid only for losses physically distinct from distributed atomic absorption
Electronics	Linear transfer functions and calibrated SQL reference	Requires measured $H_p, H_c, w, S_{\text{el}}$ at every analysis frequency

D Parameter provenance and sensitivity

The classifications below distinguish quantities measured for the cited apparatus, assumed in the reduced model, fitted or inherited phenomenologically, and algebraically derived. A dash means that the parameter does not enter that model layer. Unless an experimental uncertainty and its source are stated explicitly, a numerical “audit range” below is only an illustrative sensitivity sweep; it is not a confidence interval, prior, or measurement uncertainty.

Table 22: Measured, assumed, fitted, inherited, and derived parameters.

Parameter	Physical definition	Class and source	Procedure or allowed range	Sensitivity of G_p, G_c, S_-
$L = 12.5 \text{ mm}$	Vapor interaction length	Measured; experimental cell [1]	Use manufacturing/calibration uncertainty when available	Gains depend through $\exp(M_{cl}L)$; noise through $W(L)$ of Eq. (56)
$T = 121 \text{ }^\circ\text{C}$	Cell-reservoir operating temperature	Experimentally measured operating point [1]	Not a fitted model optimum	Strong density and Doppler dependence; must be rescanned
$P_{\text{pump}} = 600 \text{ mW}$, $P_0 = 8 \text{ } \mu\text{W}$	Incident pump and probe-seed powers used for squeezing	Experimentally measured operating point [1]	The probe-seed was approximately $10 \text{ } \mu\text{W}$ in the apparatus description; $8 \text{ } \mu\text{W}$ for the cited squeezing trace, with raw outputs $P_p = 111 \text{ } \mu\text{W}$ and $P_c = 109 \text{ } \mu\text{W}$	Pump changes the steady state and power broadens the poles (Table 7); the probe-seed sets the bright-beam linearization and the SQL
$w_{\text{pump}} = 530 \text{ } \mu\text{m}$, $w_p = 330 \text{ } \mu\text{m}$ $\theta = 0.32^\circ$	$1/e^2$ intensity radii, not diameters Phase-matching angle	Measured [1] Measured geometry [1]	Use the reported beam convention Zero-angle check required	Set Rabi frequency and spatial overlap Produces 1.38 MHz rms angular Raman broadening at $121 \text{ }^\circ\text{C}$, comparable to the power-broadened Raman pole width
$\Delta/(2\pi) = +0.90 \text{ GHz}$, $\delta/(2\pi) = -8 \text{ MHz}$	Experimental optical and two-photon operating detunings	Experimentally measured operating point [1]	Not identified with the archived numerical optimum	Corrected-sign, one-use-normalized reduced model gives $G_p = 403.535$, $G_c = 405.854$ after the thermal transit reset; the 0.74 residual is not refitted
$\Gamma/(2\pi) = 5.746 \text{ MHz}$	Natural optical linewidth used in the model	Literature benchmark	Ordinary-frequency value converted once to angular rate	Sets optical resolvents and Voigt width
$\gamma_{\text{opt}}/(2\pi) = 0.742 \text{ MHz}$ at $121 \text{ }^\circ\text{C}$	Added collisional optical half-width $\frac{1}{2}\beta N$	Derived from the density and the self-broadening coefficient	Scales linearly with $N(T)$	Broadens the optical poles of Table 7

Continued on next page

Parameter	Physical definition	Class and source	Procedure or allowed range	Sensitivity of G_p, G_c, S_-
$\sigma_v = 193.7 \text{ m/s}$ at 110°C ; 196.5 m/s at 121°C	One-axis thermal rms speed	Derived from Maxwell statistics	$\sigma_v = \sqrt{k_B T/m}$	Sets both one-photon and mismatch-projected Doppler widths (one-photon rms 243.6 MHz and 247.2 MHz ; angular-mismatch rms 1.36 MHz and 1.38 MHz , at 110°C and 121°C , respectively)
$\ell_s = 0.74$	Effective line-strength or overlap multiplier	Inherited from code; phenomenological coefficient with unknown fit provenance	Original fitting record unavailable; illustrative $\pm 10\%$ sweep $0.666\text{--}0.814$, not an uncertainty interval	At fixed $N_F = 3$ atomic response, (G_p, G_c) changes from $(188.447, 189.054)$ through $(403.535, 405.854)$ to $(857.836, 863.691)$; physical S_- remains unavailable
$\kappa = 0.1$	Coupling of absorbed pump fraction to detected pump-scattering noise	Inherited from code; assumed phenomenological coefficient	No documented fit; illustrative sweep $0\text{--}0.2$, not an uncertainty interval, pending apparatus calibration	No effect on G_p, G_c ; the diagnostic obeys $N_{ps} = \kappa(1 - \exp[-0.149609])$ with slope 0.138956 , but normalized $S_- (\Omega)$ requires F_{ps} and detector calibration
$\gamma_t/(2\pi) = 100 \text{ kHz}$ at 121°C ($\gamma_{gg,\text{total}}/(2\pi) = 102.882 \text{ kHz}$)	Thermal transit-reset input; the total ground-coherence rate also contains a 2.882 kHz collisional pure-dephasing term	Inherited from code; assumed phenomenological rate	Illustrative $\gamma_t/(2\pi)$ sweep $90\text{--}110 \text{ kHz}$, not an uncertainty interval; independent linewidth fit absent	Recomputed $N_F = 3$ endpoints give $(G_p, G_c) = (440.250, 442.733)$ at $\gamma_t/(2\pi) = 90 \text{ kHz}$, $(403.535, 405.854)$ at 100 kHz , and $(370.626, 372.790)$ at 110 kHz (total rates $92.882, 102.882$, and 112.882 kHz , respectively); the pole width is not a direct measurement of this parameter
$1 - \eta_{\text{path}} = 5.5(2)\%$	Post-cell optical-path loss	Measured [1]	Allowed range $5.3\text{--}5.7\%$; not cell absorption	No effect on gains; for symmetric loss $\partial S_{\text{det}}/\partial \eta = S_{\text{out}} - 1$
$1 - \eta_{\text{det}} = 8.0(2)\%$	Detector and remaining system inefficiency	Measured [1]	Allowed range $7.8\text{--}8.2\%$	No effect on gains; combines multiplicatively with path efficiency
$\eta_{\text{ext}} = 0.8694$	Illustrative total external efficiency 0.945×0.920	Derived from measured losses	The paper also quotes total loss $13.5(2)\%$; definitions must be reconciled	Maps S_{out} to $1 - \eta_{\text{ext}} + \eta_{\text{ext}} S_{\text{out}}$

Continued on next page

Parameter	Physical definition	Class and source	Procedure or allowed range	Sensitivity of G_p, G_c, S_-
N_F	Retained Floquet sideband order	Numerical truncation	$N_F = 3$ production default with a full-scan $N_F = 2$ comparison	$N_F = 1$ changes the historical coefficients by 13–17% and reverses the sign of their gap, without violating any positivity test
$\ell_{\text{eff}} = 0.74/12$	Residual line/overlap factor times the sublevel average	Population supplied by trace-normalized ρ_{ss} ; duplicate removed	$\ell_{\text{eff}} = 0.0616667$; see Appendix E	Representative-state coefficient test matches the AutoOD structural factor

No entry involving ℓ_s , κ , or the adopted γ_{gg} is described as exact. Their source records and any apparatus-level fitting provenance remain required metadata. The atomic solve also begins from an already-declared cell-input probe power; it contains no EOM V_π , insertion-loss, sideband-fraction, or impurity calibration. Missing EOM provenance therefore blocks an apparatus-level mapping from RF settings to that optical input and blocks any EOM-noise claim.

These missing records activate the claim gate rather than being filled by an illustrative sweep: algebraic identities and finite-model convergence may still be reported, but quantitative apparatus-level gain remains UNSUPPORTED and physical $S_-(\Omega_{\text{SA}})$ remains UNAVAILABLE. A calibration-point agreement, if later obtained, may not promote either claim without a predeclared held-out comparison.

E Dipole and population normalization ledger

Let $d_{\text{D1}} = \langle J_e | er | J_g \rangle$ be the fine-structure reduced matrix element. The archived coefficient is not the polarization-summed hyperfine strength. It is the following *AutoOD* $q = 0$ convention:

$$\mathcal{S}_0(F_g, F_e) = \sum_{m=-\min(F_g, F_e)}^{+\min(F_g, F_e)} \begin{pmatrix} F_e & 1 & F_g \\ m & 0 & -m \end{pmatrix}^2, \quad (78)$$

$$C_F^2 = (2F_g + 1)(2F_e + 1)(2J_g + 1)(2J_e + 1)(2L_g + 1)\mathcal{S}_0(F_g, F_e) \\ \times \left\{ \begin{matrix} J_g & J_e & 1 \\ F_e & F_g & I \end{matrix} \right\}^2 \left\{ \begin{matrix} L_g & L_e & 1 \\ J_e & J_g & S \end{matrix} \right\}^2. \quad (79)$$

For $I = 5/2$, $S = 1/2$, $L_g = 0$, $L_e = 1$, and $J_g = J_e = 1/2$. Then $\mathcal{S}_0 = 1/3$, which gives the stored exact values

$$C_F^2 = \{10, 35, 35, 28\}/81$$

for $(F_g, F_e) = (2, 2), (2, 3), (3, 2), (3, 3)$, respectively.

The AutoOD orbital reduced dipole and the fine-structure reduced dipole used by GABES obey $|d_{\text{orb}}|^2 = 3|d_{\text{D1}}|^2$ in this D1 convention. Therefore

$$|d_{\text{eff}}|^2 = C_F^2 |d_{\text{orb}}|^2 = 3C_F^2 |d_{\text{D1}}|^2,$$

and the dimensionless amplitude multiplier that accompanies d_{D1} is

$$a_{ge} = \sqrt{3C_F^2}. \quad (80)$$

This derivation explains the code factor but also its limitation: Eq. (79) is a $q = 0$, Zeeman-averaged absorption convention, not a universal incoherent sum over $q = 0, \pm 1$. A polarization-resolved

Liouvillian must replace it with explicit Clebsch–Gordan amplitudes and must not multiply by a_{ge} again.

The formerly archived effective factor was assembled as

$$\ell_{\text{eff}}^{\text{old}} = \ell_s \frac{p_{F=3}}{2(2I+1)} = 0.74 \times \frac{7}{144} = 0.0359722. \quad (81)$$

This double-counted the manifold population: the trace-normalized ρ_{ss} already supplies the actual, pump-modified population of the representative $F = 2, 3$ states. The corrected one-use normalization is

$$\ell_{\text{eff}} = \ell_s \frac{1}{2(2I+1)} = \frac{0.74}{12} = 0.0616667. \quad (82)$$

In a thermal reference state the response itself contributes $\rho_{FF} = p_F$, yielding the AutoOD structural coefficient $3C_F^2 p_F / [2(2I+1)]$ without a second external p_F . The residual ℓ_s remains phenomenological and must not be silently refitted to restore an old gain value.

The normalization rule for future implementations is:

1. use the fine-structure reduced dipole matrix element once;
2. use either explicit Clebsch–Gordan matrix elements or the averaged a_{ge} , never both;
3. include isotope abundance once in N ;
4. obtain ground-manifold population either from ρ_{ss} or an external p_F , never both;
5. include spontaneous-emission branching ratios in the collapse operators, not again in susceptibility amplitudes; and
6. apply external optical efficiencies only in the detection layer.

Table 23: Code-path normalization audit for the current reduced FWM solver.

Factor	Where it enters	Verdict
Reduced D1 dipole d_{D1}	Once in the macroscopic conversion in <code>gabes/observables.py</code>	no duplicate found
Hyperfine/Zee-man strength	$\sqrt{3C_F^2}$ in the weak-field drive and once in polarization readout; their product gives $3C_F^2$. It is not also present in <code>physical_coupling_norm</code> .	no algebraic duplicate found
Isotope abundance	The FWM path uses the pure- ⁸⁵ Rb density from <code>hyperfine.number_density</code> ; it does not multiply the natural 0.7217 abundance.	no duplicate in this path
Ground-manifold population	ρ_{ss} is trace normalized over the two lumped ground manifolds and supplies the population once. <code>physical_coupling_norm</code> now supplies only $1/[2(2I+1)]$.	algebraic duplicate removed; structural test
Spontaneous branching	The C_F^2 -normalized decay fractions occur in Liouvillian collapse channels; no additional branching multiplier is applied in <code>chi_phys</code> .	no duplicate found
Residual ℓ_s	Multiplies the macroscopic coupling once.	inherited, not derived

The coefficient ledger now removes the algebraic population duplication and a representative thermal-state/AutoOD coefficient identity is regression-tested. Executable tests now also inspect the pump-state ρ_{ss} and its trace-zero weak response. Those dynamical tests do not supply an independent absolute-OD scale and therefore are not, by themselves, a full weak-probe OD validation. Any future end-to-end OD comparison must retain the same one-use population-counting rule.

F Four-level model limitations and unresolved physics

The four-hyperfine-level model is suitable for qualitative gain structure, reduced hyperfine-scale intuition, and comparison with the existing reduced simulation. It is not automatically sufficient for absolute squeezing, polarization-sensitive effects, magnetic-field response, quantitative spontaneous-emission noise, or accurate Raman linewidths.

Physics outside the current basis and reservoir model includes:

- polarization-dependent Clebsch–Gordan pathways and explicit $\sigma^\pm$ and π selection rules;
- Zeeman optical pumping, ground-state orientation and alignment, dark-state formation, and magnetic-field-dependent coherences;
- interference among distinct m_F pathways and polarization-dependent spontaneous emission;
- spatial pump depletion, longitudinal attenuation, transverse Gaussian averaging, diffraction, mode mismatch, and cell-window etalons;
- buffer-gas or wall-collision dynamics, transit-time distributions, radiation trapping, and non-Markovian reservoirs;
- multidimensional velocity-changing collisions, beam angular distributions, and promotion of the separate non-collinear Raman-Doppler reference into the finite-seed segmented production path;
- a microscopic frequency-dependent atomic diffusion matrix, excess fluorescence, and atom–reservoir correlations;
- detector transfer functions, electronic cross-talk, common-mode rejection, and the apparatus-specific 2 MHz laser-source noise peak; and
- an independently measured SQL calibration and gain definition matched exactly to the theoretical modes.

None of these can be assumed small. The gain discrepancy of Table 18 and the 1.36–1.38 MHz angular Doppler scale, which is comparable to the power-broadened Raman pole width itself, are order-unity effects; the present calculation does not yet define a squeezing-bandwidth prediction.

G Implementation notes for the analytic layers

The following are the practical requirements implied by Secs. 10 and 18.

1. **Do not diagonalize M_q without a guard.** Use Eq. (62) by default. If Eq. (58) is used for interpretation, check $\text{cond}(S)$ and fall back when it exceeds a threshold; the two forms agree to 10^{-15} in the well-conditioned regime.
2. **Assert the invariants.** W must be Hermitian and positive semidefinite for any Hermitian, positive-semidefinite D , and $W(0) = 0$. These are cheap tests that catch sign and conjugation errors in the creation/annihilation bookkeeping.
3. **Treat the stationary subspace explicitly at zero frequency.** For $\Omega \neq 0$ the unique trace-mode residue vanishes by Eq. (26). At $\Omega = 0$, solve on the trace-zero subspace or use the Drazin inverse; do not evaluate a formal $0/0$ term. If more than one stationary mode exists, project the full stationary subspace and check every zero-mode residue.
4. **Track residues, not only eigenvalues.** A narrow pole with a negligible residue is not a visible spectral feature. Ranking poles by $|A_m|/\Gamma_m$ rather than by Γ_m alone identifies which ones actually shape $S_-(\Omega_{\text{SA}})$.
5. **Recompute K after any normalization change.** Equation (50) is contracted with J and is therefore basis dependent through Eq. (7).

6. **Report the analysis frequency with every spectrum.** The Ω_{SA} dependence enters both M_q and D ; a spectrum quoted without its frequency grid cannot be compared with a measurement.

H Reproducibility

The archived Floquet and velocity-grid tables and the separate Option-A literature-point transfer, Maxwell/ Q parity, commutator fixture, and model-layer comparison, plus the ℓ_s , κ , transit-floor, and two-dimensional Raman-Doppler grid/cutoff sensitivity blocks, are generated by `analysis/squeezing/analytic_reconstruction/convergence_audit.py`. Its human-readable and machine-readable outputs are `generated/convergence_audit.md` and `generated/convergence_audit.json`. Those files retain the separate historical dressed- k scans as numerical-isolation fixtures, while their literature-point Option-A section uses the current production atomic dissipator and shared Maxwell/ Q constructors. Its Floquet assembly remains analysis-owned as a cross-check, while production independently supplies general-order continued-fraction core and compiled-kernel paths. It also reconstructs M_E , Q , and T_E independently from literal SI prefactors and `scipy.linalg.expm`. At $N_F = 3$ the production/reference maximum differences are 1.64×10^{-14} or smaller. Tables 10, 16, and 17 are generated from that checked path. Passive Beer–Lambert, unrotated-to-symmetric mismatch equivalence, zero-coupling identity, constant-segment parity, the structural normalization ledger, an independent stored- χ Option-A transfer golden, Bogoliubov Q conversion, and Manley–Rowe energy bounds are pinned in `tests/test_fwm_propagation.py`. General-order continued-fraction versus dense-block parity, all-harmonic trace and residual invariants, compiled-kernel parity for $N_F = 1, 2, 3$, and the full-scan convergence gate are pinned in `tests/test_fwm_floquet.py`. The pump-only state diagnostics, static-frame/Floquet parity, finite-seed limit, projected-DC response, and direct pole/residue parity are emitted by `convergence_audit.py`; their solver-level fixtures are pinned independently in `tests/test_fwm_pump_response.py`. The lab-frequency contract, collinear and equal- k limits, analytic rms, and two-dimensional grid/cutoff checks are pinned in `tests/test_fwm_angular_doppler.py`. The detailed analytic-identity table, full pole-spectrum table, and physicality-margin table remain report derivations or separately reported checks and are not described as regenerated by the audit script. The document-level status is reported once, in Table 5 and the conclusion, rather than repeated in a second acceptance table.

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